AI based System for human Emotion detection

Final Year Project

Session 2020-2024

A project submitted in partial fulfillment of the degree of

BS in Computer Science

Department of Computer Science

Faculty of Computer Science & Information Technology

The Superior University, Lahore

Spring 2024

Type (Nature of project)	[✓] **D**evelopment [ ] **R**esearch [ ] **R&D**
Area of specialization	Web Based Application
FYP ID	FYP-BCSM-F23-037

Project Group Members

Sr.#	Reg. #	Student Name	Email ID	*Signature
(i)	Bcsm-F20-311	Sarah Mehmoood	Bcsm-f20-311@superior.edu.pk	
(ii)	Bcsm-F20-316	Rabeed Shahid	Bcsm-f20-316@superior.edu.pk	

The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to work of others

Plagiarism Free Certificate

This is to certify that, I Sarah Mahmood Daughter of Tahir Mahmood, group leader of FYP under registration no **FYP-BCSM-F23-037** at Computer Science Department, The Superior University, Lahore. I declare that my FYP report is checked by my supervisor.

Date: **07-06-2024** Group Leader: Sarah Mahmood Signature: _____

Name of Supervisor: Ms. Arshia Naeem

Designation: Senior Lecturer

Approval: _____

HOD: Dr. Irfan ud din

Approval: _____

AI based System for human Emotion detection

Change Record

Author(s)	Version	Date	Notes	Supervisor's Signature
Sarah	1.0	12-10-2023	Literature Review Gantt Chart Empathy Map	
Sarah	1.1	23-11-2023	Changes in Use case model	
Rabeed	1.2	29-12-2023	Changes in Architecture Diagram DFD	
Sarah	1.3	5-04-2024	Changes in Testing	
Rabeed	1.4	19-04-2024	Changes in chapter 7	
Sarah	1.5	17-05-2024	Removing errors	
Sarah	1.5	27-05-2024	Final version	

APPROVAL

PROJECT SUPERVISOR

Comments: _____

_____

Name:_____

Date:_____ Signature:_____

PROJECT MANAGER

Comments: _____

_____

Date:_____ Signature:_____

HEAD OF THE DEPARTMENT

Comments: _____

_____

Date:_____ Signature:_____

Table of Contents

Chapter 1 1
Introduction 1
1.1. Background 2
1.2. Motivations and Challenges 3
1.3. Goals and Objectives 4
1.4. Literature Review/Existing Solutions 5
1.5. Gap Analysis 5
1.6. Proposed Solution 6
1.7. Project Plan 6
1.7.1. Work Breakdown Structure 7
1.7.2. Roles & Responsibility Matrix 8
1.7.3. Gantt Chart 9
1.8. Empathy Map 10
Chapter 2 11
Software Requirement Specifications 11
2.1. Introduction 12
2.1.1. Purpose 12
2.1.4. Product Scope 14
2.1.5. References 14
2.2. Overall Description 15
2.2.1. Product Perspective 15
2.2.2. User Classes and Characteristics 16
2.2.3. Operating Environment 17
2.2.4. Design and Implementation Constraints 17
2.2.5. Assumptions and Dependencies 18
2.3. External Interface Requirements 19
2.3.1. User Interfaces 19
2.3.2. Hardware Interfaces 20
2.3.3. Software Interfaces 20
2.4. System Features 21
2.4.1. System Feature 1 22
2.4.1.1. Description and Priority 23
2.4.1.2. Stimulus/Response Sequences 24
2.4.1.3. Functional Requirements 24
2.4.2. System Feature 2 25
2.4.2.1. Description and Priority 25
2.4.2.2. Stimulus/Response Sequences 26
2.4.2.3. Functional Requirements 26
2.5. Other Nonfunctional Requirements 29
2.5.1. Performance Requirements 29
2.5.2. Security Requirements 30

2.5.3. useability Requirements 31
2.5.4. Reliability Requirements 32
2.5.5. Maintainability Requirements 32
2.6. Other Requirements 33
Chapter 3 36
Use Case Analysis 36
3.1. Use Case Model 37
3.2. Use Case Descriptions 38
Chapter 4 39
System Design 40
4.1. Architecture Diagram 40
4.2. Domain Model 41
4.3. Entity Relationship Diagram with data dictionary 42
4.4. Class Diagram 43
4.5. Activity Diagram 44
4.6. State Transition Diagram 45
4.7. Component Diagram 46
4.8. Data Flow diagram [only if structured approach is used - Level 0 and 1] 47,48
Chapter 5 49
Implementation 50
5.1. Important Flow Control/Pseudo codes 50
5.2. Components, Libraries, Web Services and stubs 52
5.3. Deployment Environment 53
5.4. Tools and Techniques 54
5.5. Version Control 55
Chapter 6 59
Testing and Evaluation 60
6.2. Equivalence partitioning 60
6.3. Boundary value analysis 61
6.4. Data flow testing 62
6.5. Unit testing 65
6.6. Integration testing 66
6.7. Performance testing 68
6.8. Stress Testing 69
Chapter 7 70
Summary, Conclusion and Future Enhancements 71
7.1. Project Summary 71
7.2. Achievements and Improvements 71
7.3. Critical Review 71
7.4. Lessons Learnt 72
7.5. Future Enhancements/Recommendations 72
Reference and Bibliography 75

List of Figures

Fig 1.7.1	Work Breakdown Structure	7
Fig 1.7.3	Gantt chart	9
Fig 1.8	Empathy Map	10
Fig 3.1	Use case model	36
Fig 4.1	Architecture Diagram	40
Fig 4.2	Domain Model	41
Fig 4.3	ERD	42
Fig 4.4	Class diagram	43
Fig 4.5	Activity	44
Fig 4.6	State transition diagram	45
Fig 4.7	Component diagram	46
Fig 4.8	DFD level-1	47
Fig 4.8	DFD level-0	48
Fig 5.4	Deployment environment	54
Fig 5.6.1	Login Page	56
Fig 5.6.2	Sign page	57
Fig 5.6.3	User interface	57
Fig 5.6.4	Interface	58

List of Tables

1.7.2	Roles and Responsibility Matrix	8
6.1.1	Equivalence portioning	60
6.1.2	Sign up page	61
6.1.3	Login page	61
6.2	Boundary Value Analysis	62
6.3	Data flow testing	63
6.4	Unit testing	65
6.5	Integration testing	67
6.6	Performance testing	68

Chapter 1

Introduction

Chapter 1: Introduction

Imagine a virtual assistant that not only understands your words but also senses your feelings, responding with empathy and insight. This is the vision we aim to realize through our FYP, as we unveil the concept of an AI Voice and Emotion detection system. In an era characterized by rapid technological advancement, our project delves into the significance of emotionally aware artificial intelligence, a tool that can genuinely connect with users on an emotional level. Envision a world where technology not only hears our voices but also perceives our emotions, offering a level of understanding and empathy previously reserved for human interactions.

1.1. Background

In today's digital landscape, technology is striving to understand us better, aiming not just to respond to our commands but also to perceive our feelings. This initiative drives the creation of AI voice assistants that go beyond merely understanding words – they aim to comprehend our emotions too. Imagine chatting with your device, and it not only recognizes what you're saying but also senses if you're happy, sad, or frustrated. This ambitious venture comes from incredible advancements in technology. Computers are getting smarter, especially in understanding language and recognizing patterns. They're learning to decode our expressions, tone, and choice of words to interpret emotions, much like how humans do when they talk to each other. This move isn't just about making our interactions more pleasant; it's about tailoring technology to feel more human, responding to us in ways that reflect our emotional states. Deciphering emotions accurately from how we speak or type isn't easy; it's a complex puzzle. There are also concerns about how our emotional data is handled – ensuring it's kept private and secure is crucial. Yet, this technological journey isn't just about creating clever gadgets; it's about crafting tools that can genuinely understand and support us, potentially aiding in mental health support, making customer service more empathetic, and offering a glimpse into a more emotionally aware future where technology connects with us on a deeper level.

1.2. Motivations and Challenges

I. **Enhanced User Experience**:

Emotion detection enables AI to comprehend user behavior and emotions, demonstrating an ability to empathize and share the feelings of users.

II. **Improved Communication**:

By understanding emotions, AI can facilitate more effective communication, better meeting consumer needs and expectations.

III. **Personalization:**

Customizing responses based on emotional understanding creates a more engaging and tailored user experience.

IV. **Technological Advancements**:

The use of AI for emotion detection highlights the growing capability of technology to understand human emotions.

V. **Increased User Engagement**:

Interactions with emotionally intelligent assistants can enhance user satisfaction and engagement.

VI. **Potential for Applications**:

Emotion recognition has diverse applications, including mental health, customer service, and entertainment.

Challenges:

I. Accuracy of Emotion Recognition:

Achieving precise identification and categorization of emotions from diverse inputs presents a significant technical hurdle.

II. Data Privacy and Ethics:

Handling user data, particularly emotional data, demands strict adherence to privacy standards and ethical guidelines.

III. Ambiguity in Emotional Expression:

The complexity of emotions and the varied ways individuals express them pose challenges for accurate interpretation.

IV. Cross-Modal Emotion Recognition:

Recognizing emotions accurately from both voice and text inputs is technically challenging due to the differing modalities.

V. Real-time Processing:

Delivering timely and contextually appropriate responses based on detected emotions requires efficient real-time processing capabilities.

VI. User Acceptance and Trust:

Gaining user comfort and trust in AI handling their emotional data is crucial for widespread adoption.

VII. Continuous Learning and Adaptation:

AI must continuously learn and adapt to evolving language nuances and emotional expressions to remain effective.

1.3. Goals and Objectives

I. **Emotion Detection Capabilities**: Implement algorithms and models to recognize and interpret emotions expressed by users.

II. **Improved User Experience:** Offer personalized and empathetic responses based on detected emotions, enhancing user satisfaction.

III. **Versatile Input Options:** Provide users with the choice of using voice based on their preferences or circumstances.

IV. **Accurate Emotion Recognition:** Develop models that accurately identify a range of emotions to better understand user sentiments.

V. **Real-time Responsiveness:** Enable the app to generate timely and contextually relevant responses based on detected emotions.

1.4. Literature Review/Existing Solutions

In examining the current ecosystem of voice assistant technologies and competitive offerings, it becomes evident that the Emotion Detection through Voice App occupies a distinctive and promising position. A comprehensive analysis of the existing systems and competitors reveals that our application addresses a critical gap in the market. Traditional emotion recognition systems often focus on facial expressions or rely on subjective manual assessments, while voice assistants primarily emphasize functional tasks over emotional interactions. Existing Voice Assistant system apps not have the feature of emotion detection that our app offers. Its unique ability to comprehend and respond to human emotions through voice inputs distinguishes it.

1.5. Gap Analysis

A closer look at the competition solutions and voice assistant ecosystem today reveals that the Emotion Detection by Voice App is in a unique and promising category. Our application fills a significant market need, according to a thorough review of rival products and current systems. Speech assistants prioritize functional duties above emotional exchanges, while traditional emotion detection systems tend to concentrate on facial expressions or subjective manual judgments. The emotion recognition capability that our app provides is not available in other Voice Assistant system apps. It stands out because to its exceptional capacity to understand and react to human emotions through voice inputs.

1.6. Proposed Solution

The AI Voice Assistant App represents a pioneering solution that addresses the challenges posed by the limitations of existing emotion recognition systems. Our project aims to develop an AI Voice and Emotion detection system, leveraging state-of-the-art natural language processing (NLP) and emotion recognition technologies (Ling, 2016). This system will possess the ability to not only understand and respond to users' spoken words but also accurately detect and respond to their emotional states in real-time.

1.7. Project Plan

This project aims to develop an AI Voice Emotion Detection Chabot. The project will be executed in several phases, beginning with Requirement Gathering, led by Rabeed and Sarah, who will spend 14 days to collect and document project requirements. Following this, Rabeed will take charge of the Analysis phase, dedicating 7 days to analyse these requirements. Sarah will define the system architecture in a 3 day, and spend the next 4 days in the Designing phase, designing system components. The Implementation & Development phase, will be done in to last 2-3 months, will be jointly undertaken by Rabeed and Sarah. In the Final Testing phase, Rabeed will perform comprehensive testing over 6 days. Lastly, Rabeed and Sarah will collaborate for 5-6 days to do project documentation.

1.7.1. Work Breakdown Structure

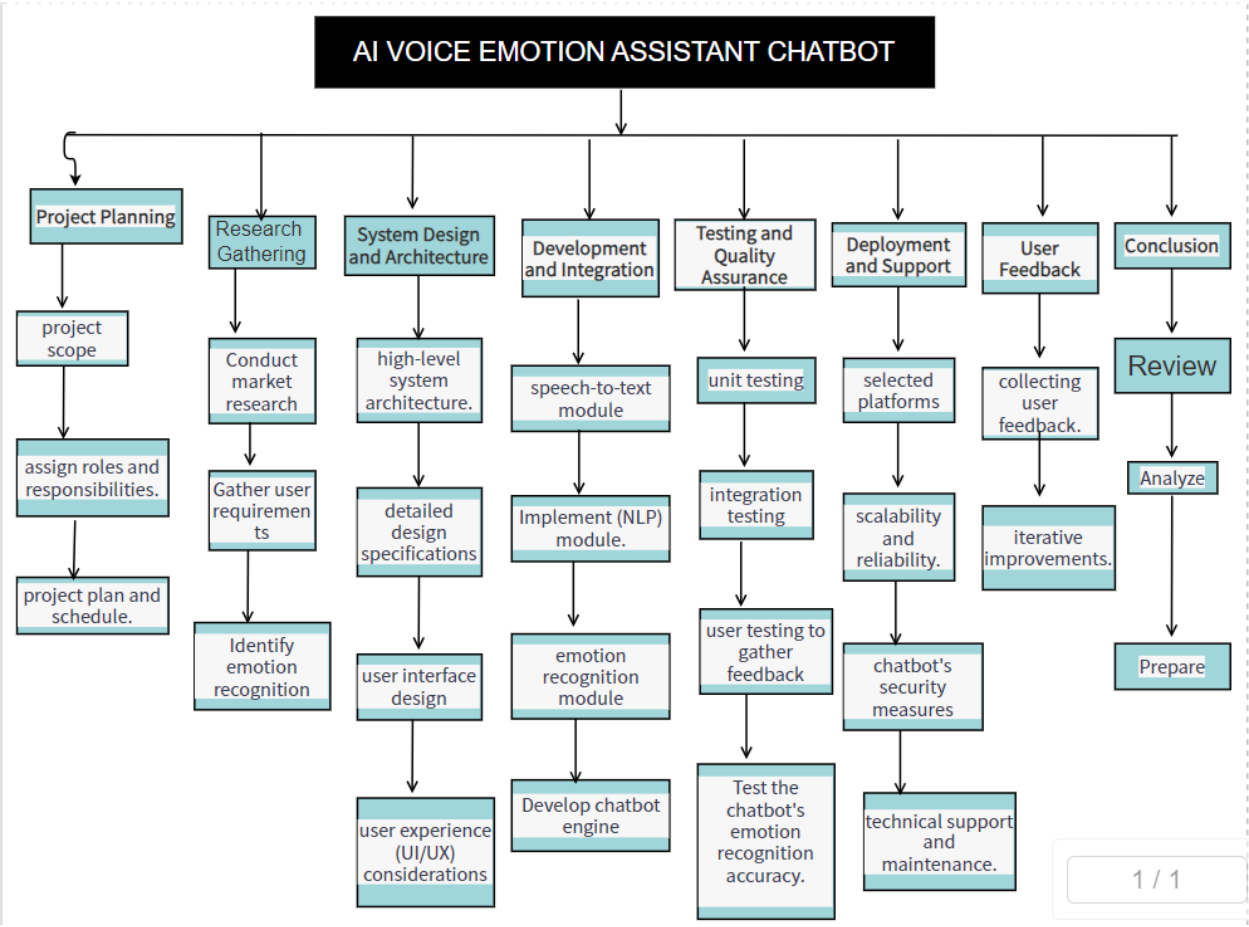


Fig 1.7.1 Work break-down Structure

1.7.2. Roles & Responsibility Matrix

WBS #	WBS Deliverable	Activity #	Activity to Complete the Deliverable	Duration (# of Days)	Responsible Team Member(s) & Role(s)
1	Requirement gathering	1		14 days	Rabeed, Sarah
2	Analysis	2		7 days	Rabeed
3	Architecture	3		4 days	Sarah
4	Designing	4		14 days	Sarah
5	Implementation &development	5		70 days	Rabeed, Sarah
6	Final testing	6		7 days	Rabeed
7	Documentation	7		7 days	Rabeed, Sarah

Table 1.7.2 Roles & Responsibility Matrix

1.7.3. Gantt Chart

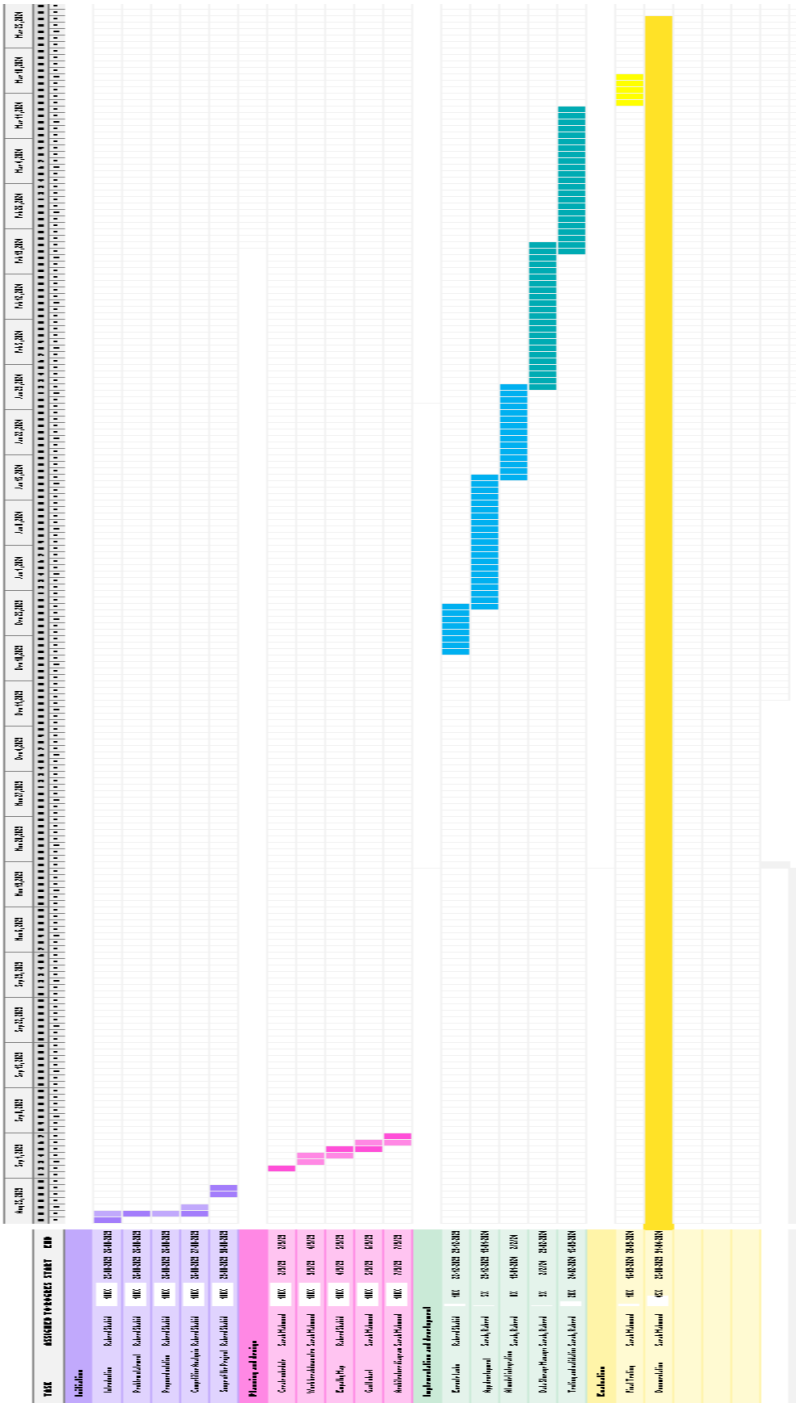


Fig 1.7.3 Gantt Chart

1.8. Empathy Map

Fig 1.8 Empathy Map

Say

- I'd love to have a tool that listens to me and figure out my emotions.
- I expect the app to respond in a friendly and empathetic manner.
- I want a tool that understand emotion and give helpful advices.

Think

- Users might think that the chat bot is a helpful and convenient tool for accessing information or emotional support.
- They may wonder how accurate the chat bot's emotion recognition is and whether it can truly understand their feelings.
- Users may ponder if the chat bot can provide effective emotional support, particularly in cases of stress, anxiety, or depression.

Do

- Users may initiate conversations with the chat bot to seek information, emotional support.
- Users may interact with the chat bot both via voice commands and text input, depending on their preferences and context.
- In situations of distress, users might reach out to the chat bot for immediate emotional support and guidance.

Feel

- Feel happiness, frustration, or comfort based on the chatbot's responses.
- Feel understood and supported when the chatbot responds correctly.
- Feel a sense of trust and comfort when the chatbot respects their privacy and data security.

Chapter 2

Software Requirement Specifications

Chapter 2: Software Requirement Specifications

2.1. Introduction

2.1.1. Purpose

The justification of this lesson is to layout the software requirements for the development of an AI based human emotion detection implemented as an system. The system will possess the capability to accurately identify a wide spectrum of emotions from users' audio and voice and respond accordingly in real-time.

2.1.2. Document Conventions

Fonts and Styles:

1. **Strong Text**: Used for headings and subheadings.
2. ***Oblique Text:*** Used to highlight key terms.
3. ***Bullets:*** Used bullets for points.

2.1.3. Intended Audience and Reading Suggestions

Intended Audience:

Developers: This Point is crucial for the development team responsible for implementing the AI based human detection system. Developers should focus on functional requirements, technical specifications, and performance criteria.

Project Managers: Project managers will find precious information in the overview sections, including the system description, features, and background.

Testers: Testers should pay close attention to the functional requirements and performance criteria to design extensive test cases. The section on non-functional requirements is especially important for testing purposes.

Users: Users, including end-users, can gain insights into the capabilities and limitations of the AI system by referring to the overview sections, such as system description and features.

Documentation Writers: Those responsible for creating user manuals and technical documentation can find related information on system functionality, user authentication, and future enhancements.

Reading Suggestions:

Overview:

- Start with the introduction to understand the purpose and scope of the document.
- Read the system overview sections for a high-level understanding of the AI Voice Assistant system.

Functional Requirements:

- Developers should focus on understanding voice recognition, emotion detection, and real-time response requirements.
- Testers can extract functional requirements to design test cases

Non-Functional Requirements:

- Developers and testers should pay attention to performance, security, and scalability requirements.
- Project managers can use this section to understand critical success factors.

Overview:

- Start with the introduction to understand the purpose and scope of the document.
- Read the system overview sections for a high-le Limitation:
- All clients ought to be mindful of confinements that might affect the system's functionality.

Future Enhancements:

- Developers and extend supervisors may discover important bits of knowledge into future highlights and improvements.

Conclusion:

- Summarize key focuses from the conclusion.
- Explore any extra data or specialized subtle elements in the reference sections. Understanding of the AI Voice Assistant system.

2.1.4. Product Scope

• **Real-time Feeling Recognition:**

Real-time feeling allows the recorder to recognize the emotions communicated by the client in the midst of their talk. This capability licenses the Voice Right hand to provide speedy and appropriately critical responses that alter with the user's excited state.

•**Voice-based Interface**

The voice-based interface is arranged for steady communication through talked lingo, apportioning with the require for text-based input. Clients emotions can be detect when the start to speak. This arrange choice ensures that client's emotions are detected properly.

•**Personalization**

Implementing machine learning algo within the system to customize responses based on individual user profiles and their emotional states. This feature ensures that interactions with the system are tailored to each user's unique preferences and emotional context. By analyzing user data and emotions, the system can provide responses that are specifically relevant and meaningful to each user, and tell them how they can change their mood or made their emotions better.

•**Versatile Integration**

It enables the voice working to easily adapt and use into different industries and used for different applications. This adaptation makes the emotion detection system a useful system for various contexts, including mental health support, customer service, crime departments, and personal assistance.

2.1.5. References

"User-Centric Design for Voice Interfaces: Enhancing Emotional Interaction"

- **Author:** Emily UX Designer
- **Published in:** Interaction Design Journal

- **Date:** August 2022

"Continuous Learning in systems: Improving Emotion Recognition Over Time"

- **Author:** AI Research Collective
- **Published in:** International Conference on Artificial Intelligence
- **Date:** October 2022

2.2. Overall Description

2.2.1. Product Perspective

This system is developed to meet the growing demand for intelligent voice interaction that goes beyond simple command recognition. Unlike traditional voice assistants, the AI Voice Assistant system places a strong emphasis on understanding and responding to users' emotional states in real-time.

Product Family:

While this product is currently a standalone entity, there is potential for it to become part of a broader product family in the future. Future enhancements may include integration with other AI-driven applications or services, contributing to a comprehensive ecosystem of intelligent and emotionally aware technologies.

Subsystem Interconnections and External Interfaces:

The AI Voice Assistant system operates as a self-contained system, interacting directly with the user through voice inputs. It relies on the following major subsystems:

- **Voice Recognition System:** Responsible for accurately transcribing and understanding users' spoken words.
- **Emotion Detection System:** Utilizes advanced machine learning models to identify a spectrum of emotions from voice inputs.
- **Response Generation System:** Formulates responses based on both the content and emotional tone of the user's input.

The external interfaces include:

User Interface (UI): The framework serves as the essential interface for clients to associate with the system.

Backend Services: The application may communicate with backend administrations for client verification, ceaseless learning, and future upgrades.

2.2.2. User Classes and Characteristics

Consistent Users:

Aspects:

- Regular users regarded the majority of the user systems.
- They have a basic to moderate level of technical expertise. But still help on their provided knowledge.

Power Users:

Aspects:

- Power users are frequent and advanced users of the system.
- They possess a high level of technical experience.
- They may use advanced features, provide feedback, and explore the full range of capabilities offered by the system.

Natural Users:

Aspects:

- Casual users interact with the system on layering levels.
- They may have varying levels of technical expertise.
- Their usage is highly demanded, and they may use the system for entertainment.

Technical Administrators:

Aspects:

- They manage and configure backend settings, perform updates, and ensure the smooth operation of the system.
- They may also be involved in monitoring and analyzing system performance.

New Users:

Aspects:

- New users are individuals who have recently started using the system.
- They may have limited familiarity with voice assistants or system interfaces.

2.2.3. Operating Environment

Hardware operating environment:

- The system is designed as a model.

Software Components:

- The system may require access to the device's microphone for voice input .
- It may utilize backend services for tasks such as user authentication.
- Integration with Html, css, bootstrap for website development**.**

Internet Connection:

- The system is designed to operate in an environment with internet connectivity.
- Certain features, such as continuous learning and updates, requires excellent internet connection.

External Working:

- This framework may interface with consent program for progressed functions to work properly or future enhancements.

Security:

- The system ought to follow the standard security practices to protect user data.

2.2.4. Design and Implementation Constraints

Corporate and Regulatory Policies:

- The development must adhere to corporate policies and regulatory guidelines concerning user data privacy and security.
- Compliance with relevant data protection laws and regulations.

Hardware Limitations:

- The system should be designed to operate within the hardware limitations of mobile devices, considering factors such as processing power, memory, and storage.

- Timing requirements for voice recognition and response generation should align with the capabilities of the target devices.

Interfaces to Other Applications:

- Integration with other applications should be considered, and the system should not interfere with the normal operation of coexisting applications on the user's device.
- Compatibility with commonly used applications or services may be explored for enhanced user experience.

Specific Technologies, Tools, and Databases:

- Development should be carried out using React Native for cross-platform compatibility.
- Specific machine learning libraries or frameworks may be required for emotion detection.
- Databases used for user authentication and continuous learning should align with organizational standards.

Design Conventions or Programming Standards:

- The development team should adhere to established design conventions and programming standards to maintain code consistency and facilitate future maintenance.

Maintenance Responsibilities:

- Clear delineation of maintenance responsibilities between the development team and the customer's organization, particularly & is responsible for ongoing support and updates.

2.2.5. Assumptions and Dependencies

Assumptions:

Availability of Internet Connection:

- Users will have consistent access to the internet for features

User's Consent for Voice Recognition:

- Users will provide explicit consent for the system to access and process voice inputs for transcription and emotion detection.

Continuous Adaptability:

- Improve emotion detection accuracy over time based on user interactions.

Dependencies:

Third-Party Machine Learning Libraries:

- The project is dependent on the availability and compatibility of frameworks for implementing emotion detection algorithms.

Backend Services:

- The system depends on backend services for user authentication, continuous learning, and future enhancements. The availability and reliability of these services are critical.

User Interface Design Guidelines:

- The project depends on adhering to user interface design guidelines to ensure a good and user-friendly experience.

2.3. External Interface Requirements

2.3.1. User Interfaces

The AI system's client interfacing are planned to give an natural and locks in involvement. With cross-platform compatibility in intellect, the client interfacing will be reliable over all platforms.

User Interface Components:

Voice Input:

Users can associated with the framework utilizing voice commands. The interface ought to consistently capture and prepare talked words.

Text Output:

Responses produced by the framework are shown in literary format.

Microphone and Speaker Controls:

Options for empowering or crippling the receiver input.

User Authentication:

If required, an verification interface guarantees secure get to to personalized highlights and data.

Responsive Design:

The client interface is outlined to be responsive, adjusting to distinctive screen sizes and resolutions for a reliable client involvement.

2.3.2. Hardware Interfaces

Supported Gadget Types:

• The framework is outlined to run on portable gadgets, counting smartphones and tablets.

• Supported stages incorporate Android gadgets (adaptation 6.0 and over)

Microphone and Speaker Interaction:

- The framework depends on the device's amplifier for capturing voice inputs from users.
- Audio yield, speaking to the system's reactions, is conveyed through the device's speaker.

Communication Protocols:

- The communication between the framework and the device's equipment components is overseen through conventions for sound input and yield.

2.3.3. Software Interfaces

Backend Services:

Components: Custom backend administrations for client confirmation and future changes.

Data Flow:

• The framework communicates with backend administrations for client confirmation amid login.

• Continuous learning includes sending anonymized voice

Voice Recognition System:

Components: Third-party voice acknowledgment libraries or services.

Data Flow:

• Voice information captured by the device's mouthpiece is sent to the voice acknowledgment framework for transcription.

Emotion Detection System:

Components: Custom or third-party feeling discovery algorithms.

Data Flow:

• Text yield from the voice acknowledgment framework is analyzed by the feeling discovery framework to recognize the user's passionate state.

• Emotion information is utilized to produce reactions that adjust with the recognized passionate tone.

Response System:

Components: Custom response generation algorithms.

2.4. System Features

1. Voice Input Processing:

Description: The framework forms voice inputs given by users.

2. User Interaction:

Description: Users start discussions by talking to the system.

Functionality:

Transcribes talked words into content for assist analysis.

3. Emotion Detection:

Description: The framework analyzes the enthusiastic tone of client inputs.

4. User Interaction:

Users express feelings through talked words.

Functionality:

Utilizes machine learning models to distinguish a range of feelings in the user's voice.

5. Response Generation:

Description: The framework defines reactions based on client inputs and recognized emotions.

6. User Interaction:

Description: Users get reactions to their inquiries or statements.

Functionality:

Generates relevantly suitable reactions, adjusting with the identified enthusiastic tone.

7. User Authentication:

Description: Guarantees secure get to to personalized highlights and data.

8. User Interaction:

Description: Users log in to get to personalized functionalities.

Functionality:

Verifies client personality through secure confirmation mechanisms.

9. Error Dealing with and Client Guidance:

Description: Guides clients in case of mistakes or uncertainties.

10. User Interaction:

Users get direction when questions are vague or blunders occur.

Functionality:

Presents supportive messages and recommendations to direct clients effectively.

11. Third-Party Integrative (Potential):

Description: Permits integration with outside administrations or APIs for expanded functionalities

12. User Interaction:

Description: Users may advantage from extra highlights or administrations through coordinates third-party components.

Functionality:

Communicates with outside administrations to bring data or perform particular assignments.

2.4.1 System Feature 1

Voice Input Processing

Description: The Voice Input Processing feature enables the AI Voice Assistant system to accurately transcribe and understand spoken words provided by users.

User Interaction: Users initiate conversations by speaking to the system.

Functionality:

- Transcribes spoken words into text for further analysis.
- Processes voice inputs to extract relevant information for generating responses.
- Initiates the communication flow within the system, serving as the entry point for user interactions.

Benefits:

- Facilitates a natural and hands-free mode of interaction.
- Allows users to communicate with the system using spoken language, enhancing accessibility and user convenience.

2.4.1.1. Description and Priority

Description: The Voice Input Processing feature involves the accurate transcription of spoken words by users, serving as the initial step in user interactions with the AI Voice Assistant system. This feature ensures that user inputs are converted into a format that can be further analyzed and understood by the system.

Priority:

High Priority

Benefit: 8

Penalty: 3

Cost: 6

Risk: 7

Benefit: The accurate processing of voice inputs is crucial for delivering a seamless and natural user experience, aligning with the system's primary mode of interaction.

Penalty: Inaccurate or inefficient voice processing can result in user frustration and a degraded user experience.

Cost: While development efforts are required to implement and optimize voice input processing, the benefits in terms of user satisfaction and engagement justify the associated costs.

Risk: The risk of prioritizing this feature as high is relatively low, given its fundamental role in the system's functionality and user engagement. However, efficient implementation and continuous improvement are essential to mitigate potential risks associated with voice processing challenges.

2.4.1.2. Stimulus/Response Sequences

User Initiates Voice Interaction:

Stimulus: User activates the system and starts speaking.

Response: The system captures the voice input and begins processing.

Voice Transcription:

Stimulus: User provides spoken input or asks a question.

Response: The system transcribes the spoken words into text.

Input Analysis:

Stimulus: Transcribed text is obtained.

Response: The system analyzes the text to extract relevant information and understand the user's intent.

Feedback to User:

Stimulus: Completion of response generation.

Response: The system provides a textual response or performs an action based on the user's input.

2.4.1.3. Functional Requirements

- REQ-SF1-1: Voice Input Capture
- REQ-SF1-2: Voice Transcription
- REQ-SF1-3: Text Data Processing
- REQ-SF1-4: Emotion Analysis Trigger
- REQ-SF1-5: Error Handling - Voice Input
- REQ-SF1-6: Real-Time Processing
- REQ-SF1-7: User Guidance

- REQ-SF1-8: Integration with Voice Recognition System
- REQ-SF1-9: Feedback to User

2.4.2. System Feature 2

Emotion Detection

Description: The Emotion Detection feature enables the AI Voice Assistant system to analyze the emotional tone expressed by users through their spoken words.

User Interaction: Users express emotions through spoken language.

Functionality:

Utilizes machine learning models to identify a spectrum of emotions in the user's voice.

Enhances the system's responses by considering the detected emotional tone.

Benefits:

Provides a personalized and emotionally aware interaction.

Enables the system to respond empathetically, improving user engagement and satisfaction.

Top of Form

2.4.2.1. Description and Priority

Description: The Emotion Detection feature involves the analysis of the emotional tone in user-spoken words, allowing the AI Voice Assistant system to respond in a manner aligned with the user's expressed emotions.

Priority:

High Priority

Benefit: 9

Penalty: 4

Cost: 7

Risk: 8

Benefit: Emotion detection significantly enhances the system's ability to understand and respond appropriately, fostering a more personalized and engaging user experience.

Penalty: Inaccurate emotion detection may lead to responses that are misaligned with the user's emotional state, impacting user satisfaction.

Cost: Developing and maintaining reliable emotion detection models incurs costs, but the benefits in user engagement justify the investment.

Risk: There is a moderate risk associated with the accuracy of emotion detection algorithms, requiring continuous improvement and adaptation to diverse user expressions.

2.4.2.2. Stimulus/Response Sequences

User Expresses Emotion:

- **Stimulus:** User communicates with the system, expressing emotions through spoken words.
- **Response:** The system captures and analyzes the emotional tone of the user's voice.

Emotion Analysis:

- **Stimulus:** Completion of user communication.
- **Response:** The system uses machine learning models to analyze the emotional tone, identifying specific emotions such as happiness, sadness, anger, etc.

User Feedback or Continuation:

- **Stimulus:** User reacts or provides additional input.
- **Response:** The system adjusts its responses based on new emotional cues, facilitating a dynamic and emotionally aware conversation.

2.4.2.3. Functional Requirements

REQ-SF2-1: Voice Emotion Analysis

The system must utilize machine learning models to perform real-time analysis of the emotional tone expressed in user-spoken words.

REQ-SF2-2: Emotion Recognition

The system must accurately recognize a spectrum of emotions, including but not limited to happiness, sadness, anger, and neutrality.

REQ-SF2-3: Contextual Emotion Integration

The system must integrate the detected emotional context into the response generation process to formulate contextually appropriate and emotionally aligned responses.

REQ-SF2-4: Continuous Learning - Emotion Models

If enabled, the system must capture anonymized emotion data for continuous learning to improve the accuracy of emotion detection models.

REQ-SF2-5: User Feedback Handling

The system must adapt its responses based on user feedback or new emotional cues provided during the conversation.

REQ-SF2-6: Error Handling - Emotion Analysis

In cases where emotion analysis encounters challenges or errors, the system must provide clear error messages and continue the conversation without relying heavily on the emotion context.

REQ-SF2-8: Security and Privacy

The system must ensure the privacy and security of emotion data, implementing measures to protect sensitive information during analysis and storage.

Real-time Emotion-based Content Recommendations

2.4.3.1. Description and Priority

This feature enables the AI Voice Assistant system to dynamically recommend content based on the user's real-time emotional state. The system prioritizes this functionality as it enhances user engagement by tailoring content suggestions to the user's current emotional context.

2.4.3.2. Stimulus/Response Sequences

User Speaks: The user initiates a conversation or query.

Emotion Analysis: The system analyzes the user's emotional state during the interaction.

Content Recommendation: Based on the detected emotion, the system generates real-time content recommendations.

User Confirmation: The user receives content suggestions and can confirm or request alternatives.

Content Delivery: The chosen content is delivered to the user.

2.4.3.3. Functional Requirements

REQ-SF3-1: Implement a real-time emotion analysis component.

REQ-SF3-2: Integrate with a content recommendation engine.

REQ-SF3-3: Ensure the system dynamically adjusts recommendations based on changing emotional states.

REQ-SF3-4: Provide users with the option to confirm or request alternative content.

REQ-SF3-5: Optimize content delivery for a seamless user experience.

This feature enhances user satisfaction by delivering personalized and emotionally resonant content recommendations, contributing to the overall effectiveness of the AI Voice Assistant system.

2.4.4 Multi-Language Support

2.4.4.1. Description and Priority

This feature enables the AI Voice Assistant system to understand and respond to users in multiple languages, enhancing its accessibility and user reach. Given the diverse linguistic landscape, this functionality is deemed a high priority to cater to a broader user demographic.

2.4.4.2. Stimulus/Response Sequences

User Language Selection: Users specify their preferred language.

Language Detection: The system detects the user's spoken language.

Multilingual NLU Processing: Natural Language Understanding processes user input in the detected language.

Response Generation: The system generates responses in the user's selected language.

Content Delivery: Responses are delivered in the chosen language through the mobile app.

2.4.4.3. Functional Requirements

REQ-SF4-1: Implement language detection capability during user interactions.

REQ-SF4-2: Integrate multilingual support within the Natural Language Understanding module.

REQ-SF4-3: Incorporate a language selection option in the mobile app.

REQ-SF4-4: Develop a language-specific response generation mechanism.

REQ-SF4-5: Ensure seamless content delivery in the user's chosen language.

2.5. Other Nonfunctional Requirements

2.5.1. Performance Requirements

Voice Processing:

- **Requirement:** The system must process voice inputs in real-time, with a maximum delay of 1 second from the initiation of voice capture to response generation.
- **Rationale:** Real-time processing ensures a seamless and natural interaction, maintaining user engagement and satisfaction.

Scalability:

- **Requirement:** The system must be scalable to handle a minimum of 10,000 concurrent users.
- **Rationale:** Scalability is crucial to accommodate varying levels of user demand, ensuring consistent performance under high load conditions.

Emotion Analysis Response Time:

- **Requirement:** Emotion analysis must be performed within 500 milliseconds from the completion of voice input processing.
- **Rationale:** Quick emotion analysis contributes to the generation of timely and emotionally aligned responses.

Error Handling Response Time:

- **Requirement:** The system must respond to errors and provide user guidance within 3 seconds from the identification of an error condition.
- **Rationale:** Quick error response enhances user understanding and reduces frustration.

Security Measures:

- **Requirement:** The overhead introduced by security measures, including encryption and secure data transmission, must not exceed 5% of the overall processing time.
- **Rationale:** Balancing security measures with performance is crucial to maintain efficient system operation.

2.5.2. Security Requirements

Voice Input Monitoring:

- **Requirement:** The system must include a monitoring mechanism to detect and prevent misuse of voice input functionalities, such as the use of harmful or offensive language.
- **Safeguard:** Implement real-time content filtering to identify and block inappropriate content.

Data Security and Privacy:

- **Requirement:** The system must adhere to data protection regulations, ensuring the security and privacy of user data, including voice inputs and emotion analysis results.
- **Safeguard:** Employ encryption for data transmission and storage, and regularly conduct security audits to identify and address potential vulnerabilities.

User Authentication Security:

- **Requirement:** User authentication mechanisms must meet industry standards for security to prevent unauthorized access.
- **Safeguard:** Implement secure authentication protocols, such as multi-factor authentication, to enhance user account protection.

Error Handling Safety Measures:

- **Requirement:** Error messages presented to users must be informative and avoid revealing sensitive information.
- **Prevention:** Regularly review and update error messages to ensure they do not inadvertently disclose confidential information.

Emergency Stop Functionality:

- **Requirement:** Implement an emergency stop functionality that allows users to quickly halt voice interactions in case of unexpected or emergency situations.
- **Safeguard:** Clearly communicate the emergency stop procedure to users, ensuring it is easily accessible within the application interface.

2.5.3. Usability Requirements

Voice Input Accuracy:

Requirement: The system must achieve a minimum accuracy of 95% in transcribing voice inputs into text.

Evaluation: Conduct regular testing to assess the accuracy of voice input processing under diverse conditions and accents.

Response Clarity and Conciseness:

Requirement: Responses generated by the system must be clear, concise, and easily understandable by users.

Evaluation: Obtain user feedback through surveys and user testing to assess the clarity and effectiveness of responses.

Error Message Clarity:

Requirement: Error messages must be informative, clearly stating the issue and providing guidance for resolution.

Evaluation: Conduct usability testing to ensure that users can easily understand and act upon error messages.

User Guidance and Help Features:

Requirement: The application must include user guidance features and help documentation accessible within the interface.

Evaluation: Assess the effectiveness of user guidance features through user feedback and usability testing.

Accessibility Features:

Requirement: The application must include accessibility features, such as screen reader compatibility and text-to-speech functionality, to cater to users with disabilities.

Evaluation: Regularly test and update accessibility features based on evolving standards and user feedback.

12.2.4 Reliability Requirements

System Uptime:

- **Requirement:** The system must achieve a minimum uptime of 99.9% over any given month, excluding scheduled maintenance periods.
- **Evaluation:** Regularly monitor system uptime and address any unexpected downtimes promptly.

Error Handling and Recovery:

- **Requirement:** The system must be capable of detecting errors and recovering gracefully to prevent service disruptions.
- **Evaluation:** Conduct stress testing and simulate error scenarios to assess the system's ability to handle and recover from errors.

Data Backup and Recovery:

- **Requirement:** Regular automated backups of user data must be performed, and a reliable data recovery mechanism must be in place.
- **Evaluation:** Periodically test data recovery processes to ensure the system can efficiently restore user data.

12.2.5 Maintainability/Supportability Requirements

Documentation:

- **Requirement:** Comprehensive documentation must be maintained, covering system architecture, codebase, APIs, and user manuals.
- **Supportability:** Facilitates efficient troubleshooting, debugging, and onboarding of new developers or support personnel.

Automated Testing Suite:

- **Requirement:** The system must have an automated testing suite covering unit tests, integration tests, and end-to-end tests.
- **Maintainability:** Allows for rapid identification of issues during development and supports future modifications without introducing regressions.

Version Control:

- **Requirement:** The code base must be managed using version control systems, such as Git, to track changes and facilitate collaboration.
- **Maintainability:** Enables easy rollback to previous versions, branch management, and collaborative development.

Code Review Processes:

- **Requirement:** Establish formal code review processes to ensure code quality, adherence to coding standards, and knowledge sharing among development team members.
- **Maintainability:** Improves code quality, reduces technical debt, and enhances overall maintainability.

12.2.6 Portability Requirements

Cross-Platform Compatibility:

- **Requirement:** The mobile application must be compatible with Android platforms, ensuring a consistent user experience.
- **Evaluation:** Conduct testing on various devices and operating system versions to verify cross-platform compatibility.

Browser Compatibility:

- **Requirement:** If a web-based component is included, ensure compatibility with major web browsers (e.g., Chrome, Firefox, Safari).
- **Evaluation:** Regularly test and update browser compatibility to accommodate evolving browser standards.

Offline Functionality:

- **Requirement:** The application should provide essential functionalities even in offline mode, ensuring users can access certain features without a continuous internet connection.
- **Evaluation:** Test offline capabilities under various scenarios to verify reliable offline functionality

12.2.7 Efficiency Requirements

Response Time:

- **Requirement**: The system must generate responses to user inputs within 2 seconds on average, providing a responsive user experience.
- **Evaluation:** Monitor response times under various load conditions and optimize system performance accordingly.

Voice Input Processing Time:

- **Requirement:** Voice input processing, including transcription and initial analysis, must be completed within 1 second on average.
- **Evaluation:** Regularly assess voice input processing times and optimize algorithms for efficiency.

Emotion Analysis Speed:

- **Requirement:** The system must analyze the emotional tone of user inputs within 500 milliseconds on average.
- **Evaluation:** Conduct performance testing to ensure real-time emotion analysis without compromising accuracy.

12.3 Domain Requirements

Database Requirements:

- **Requirement:** The system must utilize a secure and scalable database to store user profiles, interaction history, and configuration settings.
- **Implementation:** Select a relational or NoSQL database based on performance and scalability requirements. Implement proper data indexing and encryption mechanisms.

User Authentication and Authorization:

- **Requirement:** Implement secure user authentication mechanisms, such as multi-factor authentication, to protect user accounts from unauthorized access.
- **Implementation:** Employ industry-standard authentication protocols and regularly update security measures to address emerging threats.

Continuous Learning Ethical Guidelines:

- **Requirement:** Define ethical guidelines for the continuous learning process, ensuring responsible and unbiased use of user data for model improvement.
- **Implementation:** Regularly review and update ethical guidelines, and involve diverse stakeholders in decision-making processes.

Chapter 3

Use Case Analysis

Chapter 3: Use case Analysis

3.1. Use Case Model

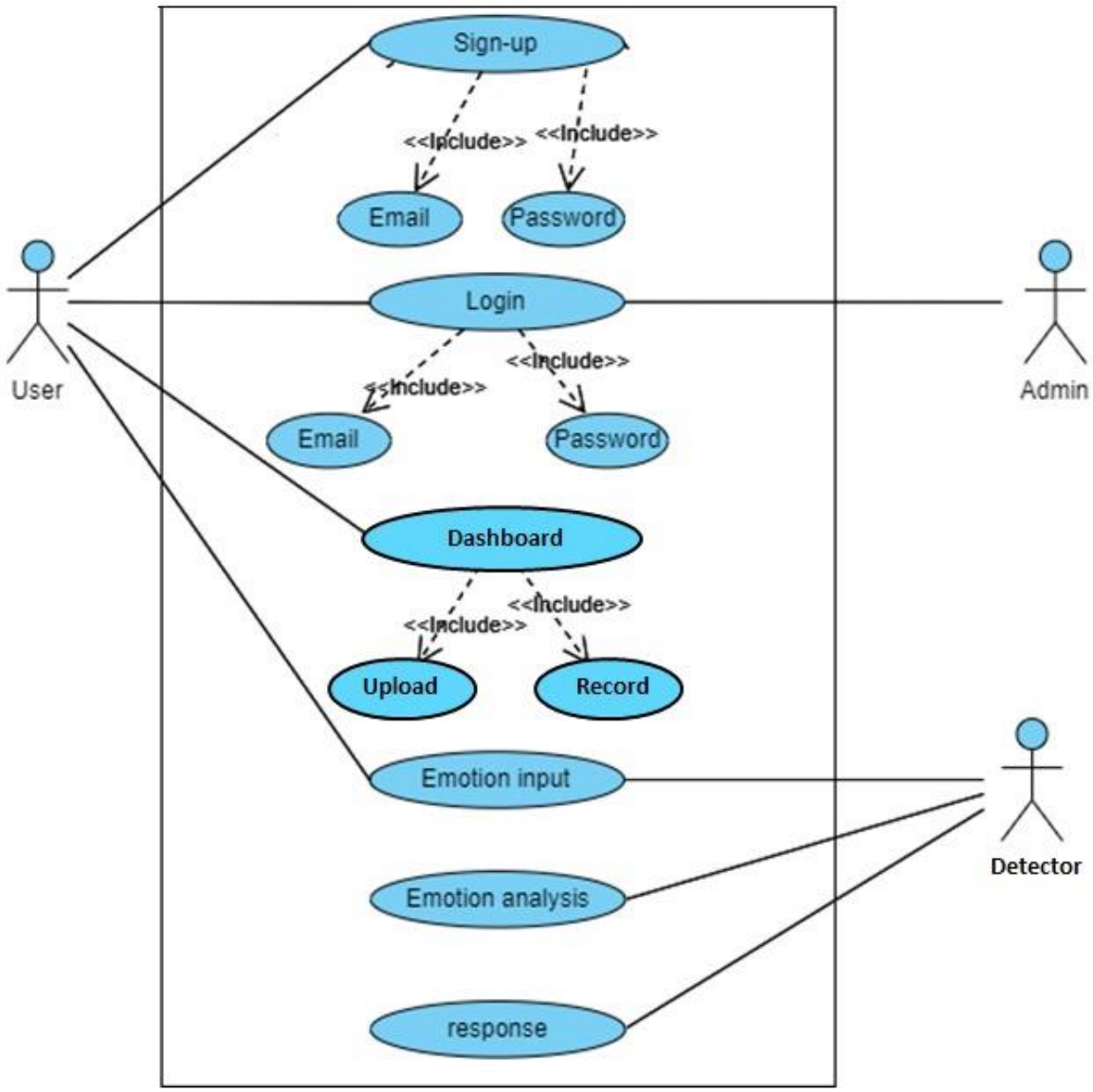


Fig 3.1 Use Case Diagram

3.2. Use Case Descriptions

1. User Initiated Interaction:

Actor: User

Description: Speaking, the user engages the system on their own.
Flow:

- User speaks a command or question.
- Voice Input captures the audio.
- The user's emotional state determines by emotion analysis.
- Natural Language Understanding processes the words spoken by user.
- The response is delivered to user.

2. Real-time Emotion-based Content Recommendations:

Actor: User, Content Recommendation Engine

Description: it can detects the emotions of user by real-time recording of voice.

Flow:

- User initiates interaction.
- User confirms or requests alternative recommendations.
- Content is delivered to the user.

3. Multi-Language Support:

Actor: User, Language Selection Module

Description: Users can interact with the system in different languages and different voices of gender.

Flow:

- User selects their preferred language.
- Language Detection identifies the spoken language.
- The natural language understanding (NLU) processes the user's input in the identified language.
- Response Generation generates responses in the chosen language.
- Responses are delivered in the selected language.

Chapter 4

System Design

Chapter 4: System Design

The System Design chapter provides a comprehensive overview of the architecture and structure of the AI Voice Assistant system. It delves into the detailed design choices made to fulfill the specified requirements, encompassing the organization of major components, data flows, and the interaction between modules. This chapter serves as a blueprint for developers, outlining the technical foundation and implementation strategies that ensure the seamless integration of voice processing, emotion analysis.

4.1. Architecture Diagram

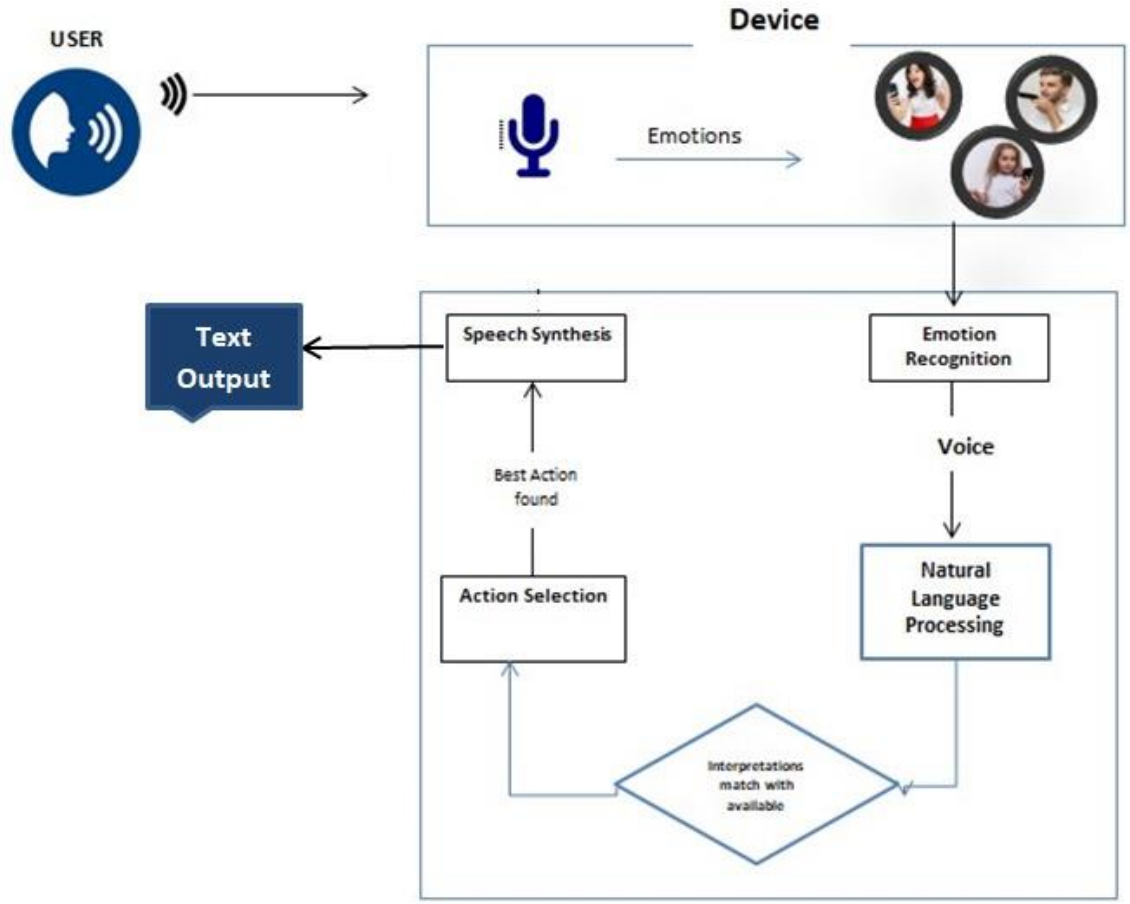


Fig 4.1 Architecture Diagram

4.2. Domain Model

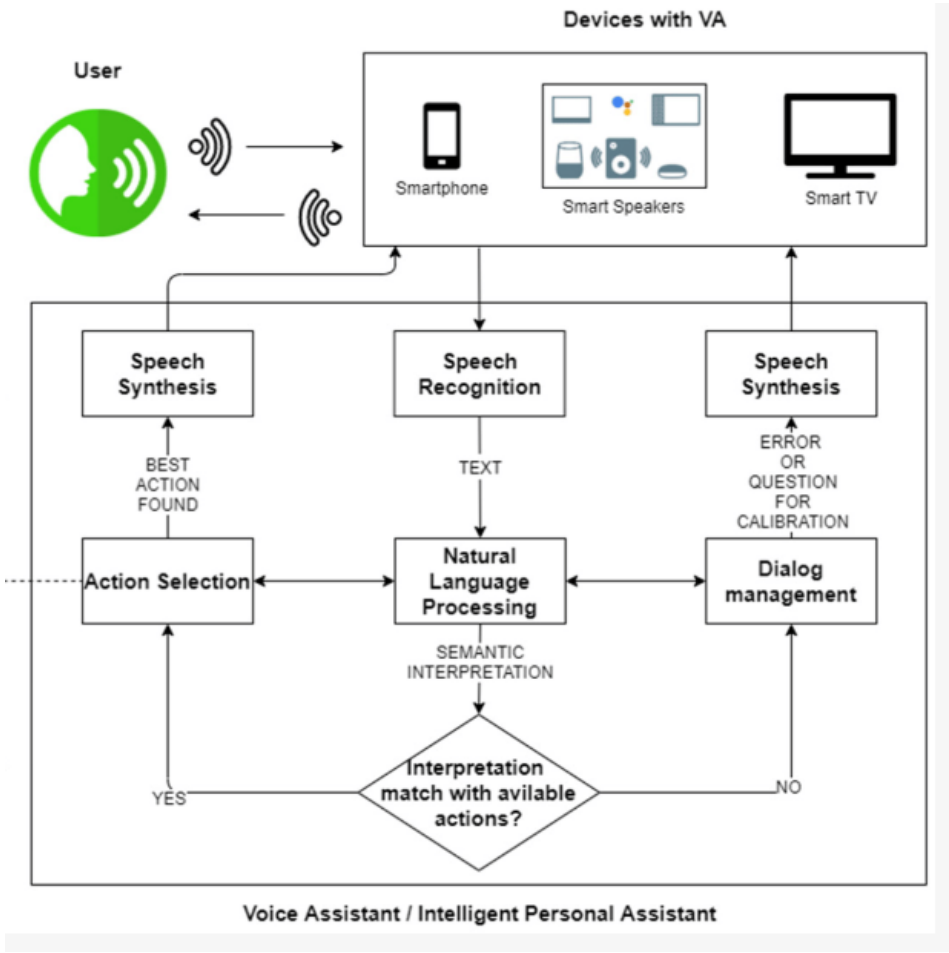


Fig 4.2 Domain Model

4.3. Entity Relationship Diagram with data dictionary

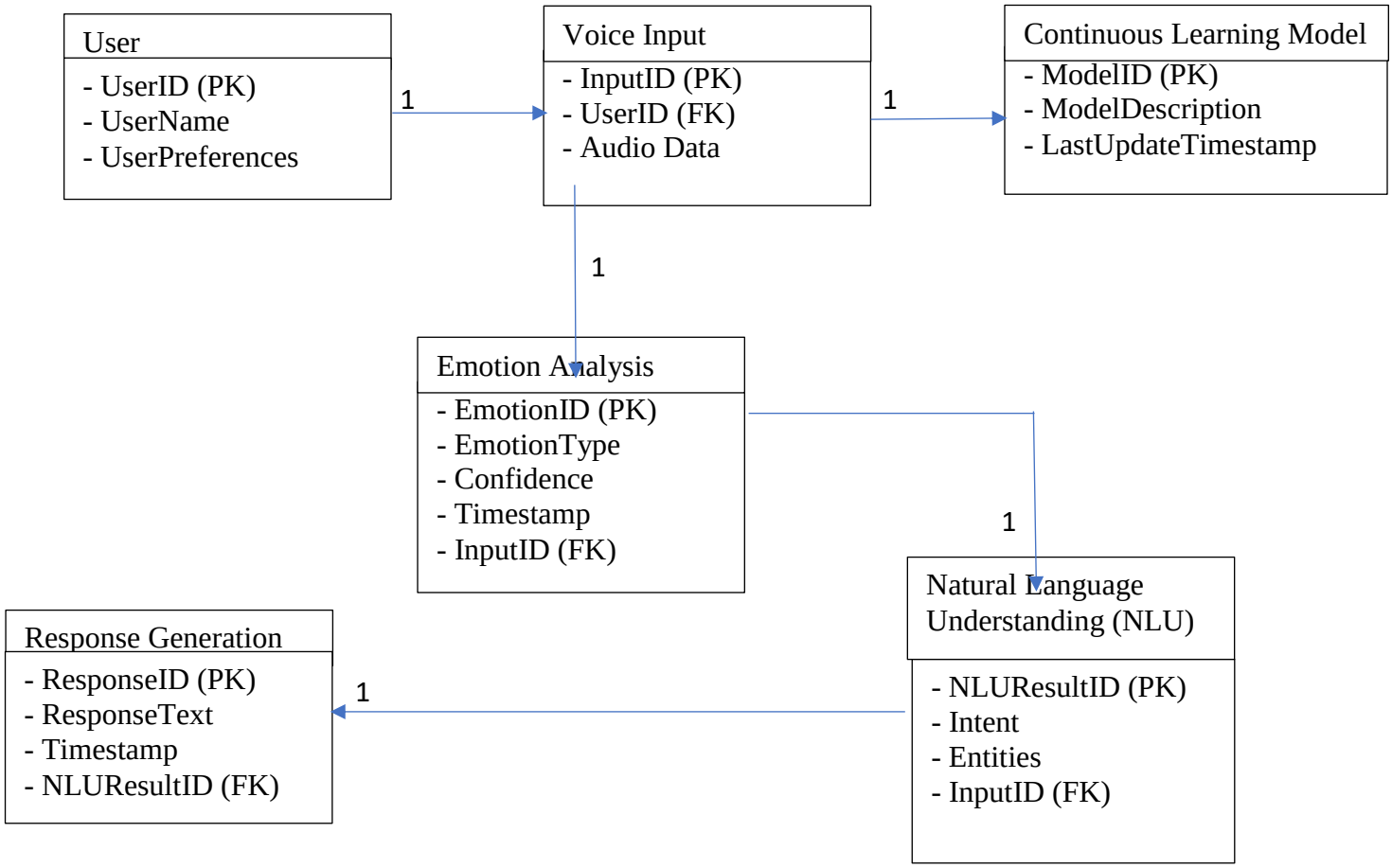


Fig 4.3 ERD Diagram

4.4. Class Diagram

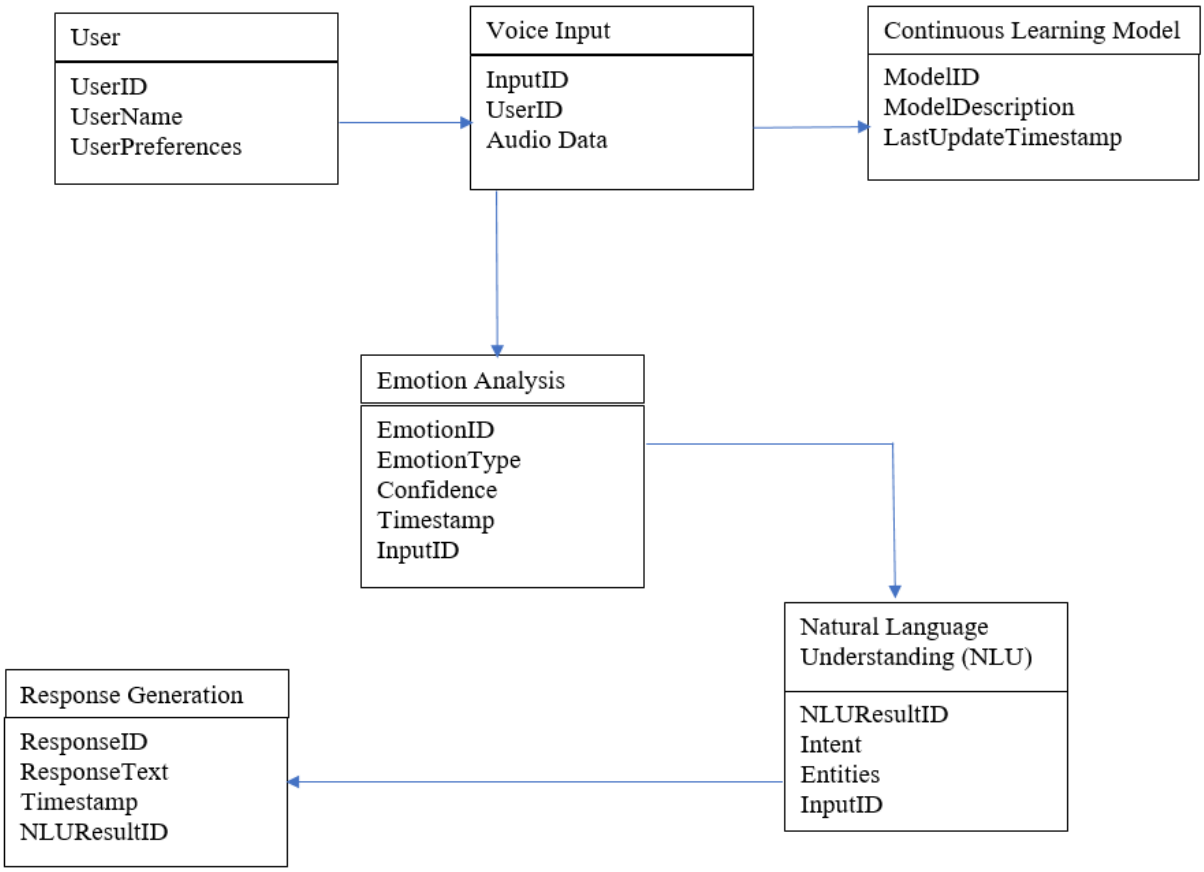


Fig 4.4 Class Diagram

4.5. Activity Diagram

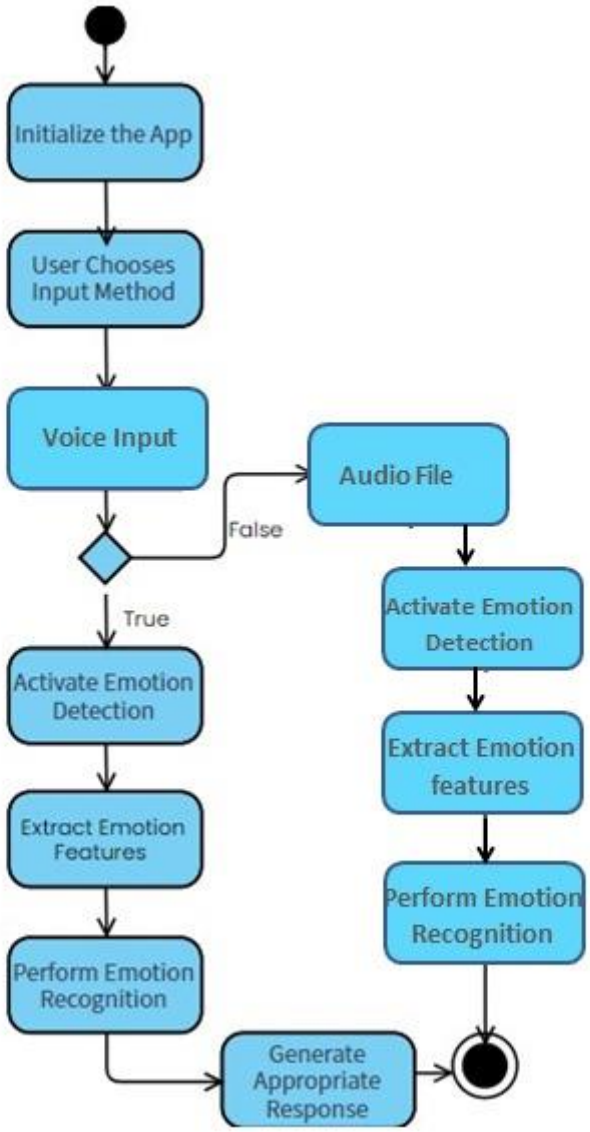


Fig 4.5 Activity Diagram

4.6. State Transition Diagram

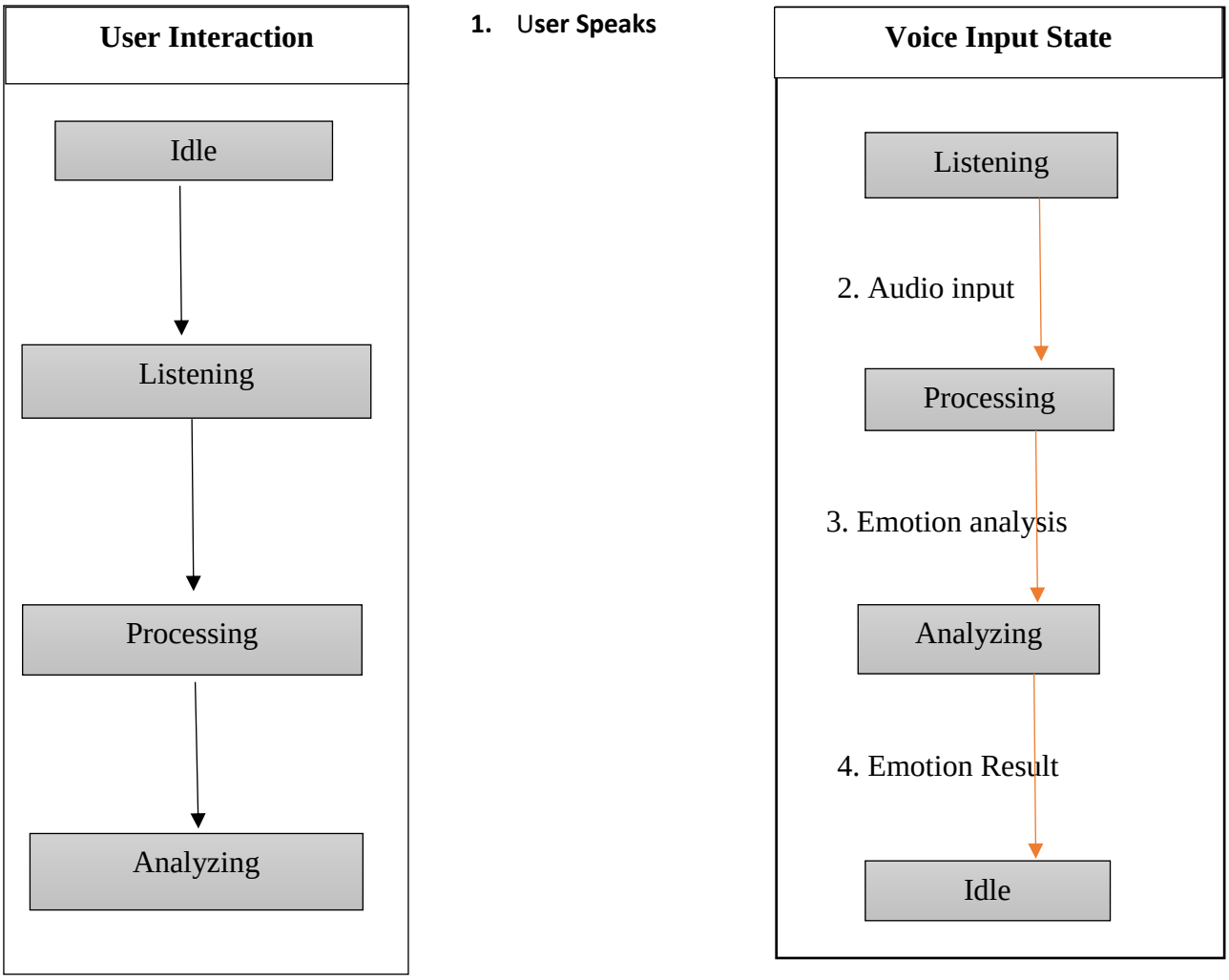


Fig 4.6 State transition Diagram

4.7. Component Diagram

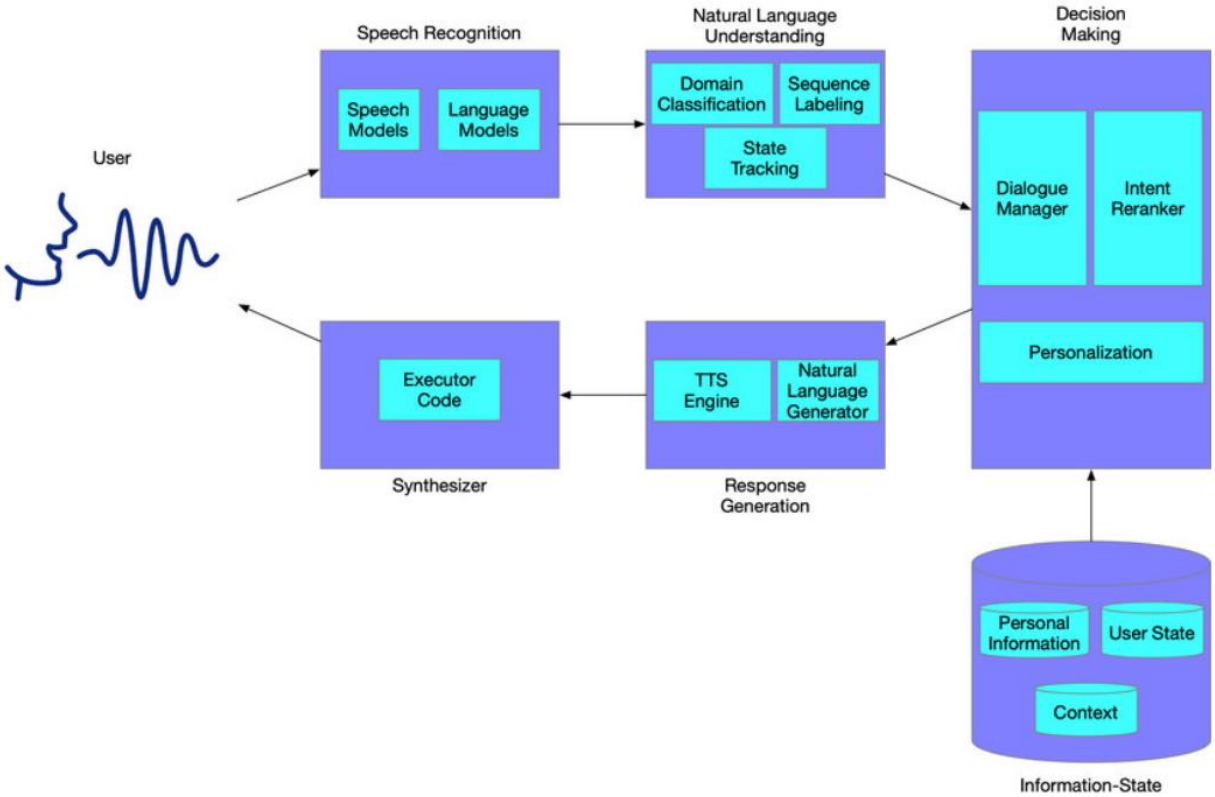


Fig 4.7 Component Diagram

4.8. Data Flow diagram

DFD-Level 1:

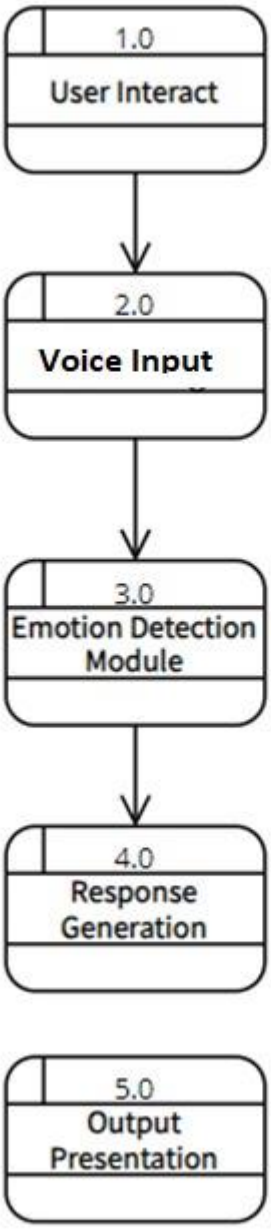


Fig 4.8 Data Flow diagram

DFD-Level 0:

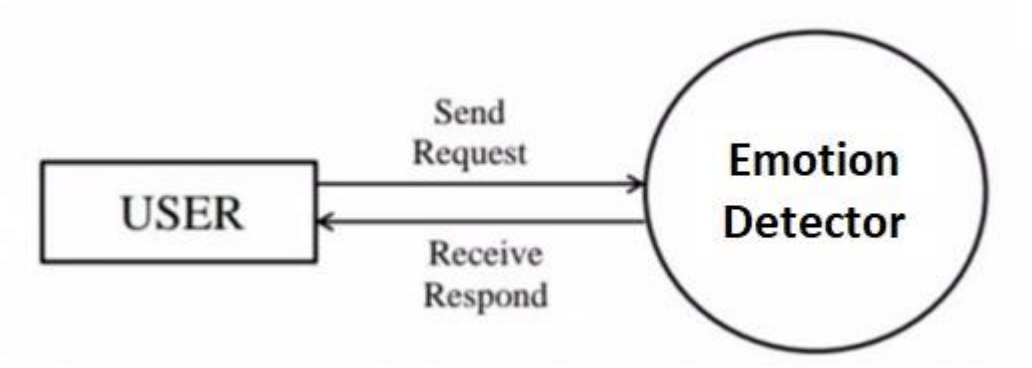


Fig 4.8 DFD level-0

Chapter 5

Implementation

Chapter 5: Implementation

The Implementation chapter outlines the actual coding and development process of the AI Voice Assistant system. It provides a detailed account of how the previously defined system design, components, and features are translated into executable code. This chapter addresses the technologies, programming languages, and frameworks utilized, ensuring alignment with the Software Requirement Specifications.

5.1. Important Flow Control/Pseudo codes

```
import React, { useState, useEffect } from 'react';
import { Text, View, Button } from 'react-native';
import * as Speech from 'expo-speech';
import { Audio } from 'expo-av';
import * as Permissions from 'expo-permissions';

export default function VoiceAssistant() {
  const [isSpeaking, setIsSpeaking] = useState(false);
  const [assistantResponse, setAssistantResponse] = useState('');

  useEffect(() => {
    askForPermissions();
  }, []);

  const askForPermissions = async () => {
    const { status } = await Permissions.askAsync(Permissions.AUDIO_RECORDING);
    if (status !== 'granted') {
      alert('Permission to access microphone denied!');
    }
```

```
  };

  const speak = async (text) => {
    setIsSpeaking(true);
    await Speech.speak(text);
    setIsSpeaking(false);
  };

  const handleVoiceInput = async () => {
    const { sound } = await
Audio.Recording.createAsync(Audio.RECORDING_OPTIONS_PRESET_HIGH_QUALITY);
    try {
      await sound.setCallback({
        onFinish: async () => {
          setIsSpeaking(true);
          const recognizedText = await Speech.recognizeAsync();
          setAssistantResponse(recognizedText);
          setIsSpeaking(false);
        }
      });
      await sound.startAsync();
    } catch (error) {
      console.error('Recording error:', error);
    }
  };

  return (
    <View style={{ flex: 1, justifyContent: 'center', alignItems: 'center' }}>
      <Button title="Press to Speak" onPress={handleVoiceInput} disabled={isSpeaking} />
```

```
    {assistantResponse !== '' && (
      <View style={{ marginTop: 20 }}>
        <Text>{assistantResponse}</Text>
        <Button title="Speak" onPress={() => speak(assistantResponse)} disabled={isSpeaking} />
      </View>
    )}
  </View>
 );
}
```

5.2. Components, Libraries, Web Services and stubs

Continuous Learning Module:

Component: Tensor Flow, CNN Sequential model

Purpose: Facilitates continuous learning by updating and improving the underlying models based on user interactions.

User Profile Management:

Component: Firebase Authentication

Purpose: Manages user authentication and maintains user profiles.

Database System:

Component: Firebase

Purpose: store user login information

Website Development Framework:

Framework: html, css, JavaScript

Purpose: Django is a high-level Python web framework designed to enable rapid development of secure and maintainable websites.

5.3. Deployment Environment

Server Environment:

Backend server components, responsible for handling user requests, emotion analysis, and natural language processing, will be deployed using containerization technologies like Docker to ensure consistency across different environments.

Database Management:

The firebase database, will be hosted in a cloud-based database service for efficient data storage and retrieval.

Micro services Architecture:

The deployment environment will adopt micro services architecture, allowing modular and independent deployment of components, ensuring flexibility, and facilitating future updates and enhancements.

Load Balancing:

Load balancing mechanisms will be implemented to distribute incoming user requests evenly across multiple server instances, optimizing system performance and preventing resource bottlenecks.

Continuous Integration/Continuous Deployment (CI/CD):

CI/CD pipelines, facilitated by tools like Jenkins, will automate the testing, building, and deployment processes, ensuring efficient and error-free updates to the production environment.

```python
app.py > ...
from flask import Flask, render_template, request, redirect, url_for, session, jsonify
import numpy as np
import librosa
import tensorflow as tf
import os
import firebase_admin
from firebase_admin import credentials, auth

app = Flask(__name__)
app.secret_key = 'supersecretkey'

# Firebase configuration
cred = credentials.Certificate(r'C:\Users\PTS\PycharmProjects\VoiceAssitant- fyp\VoiceAssitant\voiemo.json')
firebase_admin.initialize_app(cred)

# Load pre-trained emotion model
model = tf.keras.models.load_model('model/cnn_seq.h5')
model_input_shape = model.input_shape
print(f"Model input shape: {model_input_shape}")

UPLOAD_FOLDER = 'uploads'
if not os.path.exists(UPLOAD_FOLDER):
    os.makedirs(UPLOAD_FOLDER)
app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER

def extract_features(data, sr=44100, n_mfcc=40, chroma=True, mel=True):
    stft = np.abs(librosa.stft(data)) if chroma else None
    result = np.array([])
```

Figure 5.4 Deployment Environment

5.4. Tools and Techniques

Integrated Development Environment (IDE):

- **Tool:** Visual Studio Code
- **Purpose:** Provides a feature-rich and extensible environment for coding, debugging, and version control.

Programming Languages:

- **Languages:** Python
- **Purpose:** Utilized for frontend website development and backend server-side logic, respectively.

Collaboration and Communication:

- **Tool:** Slack
- **Purpose:** Facilitates real-time communication among development teams and project stakeholders.

Testing Frameworks:

- **Frameworks:** html, css, JavaScript

- **Purpose:** Enables automated testing of frontend and backend components to ensure code reliability.

5.5. Version Control

❖ Version control is a major concept in the development and management processes of the AI Voice Assistant system. The Git version control system is the platform for the project, and the repository will be centrally hosted on GitHub. This project has the following benefits :

❖ **Collaborative Development:**

Allowing for several developers to run concurrently together and separately on various aspects of the project. Each developer can create their own branches for feature development or bug fixes, and changes can be merged seamlessly.

Change Tracking:

With Git, the traceability of alterations to the code is very comprehensive. Every single commit in the code repository captures a full picture of the code at a set in time, it has been easier to detect who and when a certain change was made.

❖ **Branching Strategy:**

A very proper strategy Tok, for example, a single root allows for the separation of elements such as features, releases, and hotfixes. Such a systematic way helps to make development, testing, and deployment work autonomously and effectively.

❖ **Rollback Capability:**

Using Git, If the codebase mistakenly has any issues or bugs, then it becomes simple, and Git enables all developers to easily go back to a former, unrestored state of the codebase. Such technology, with the help of which the project can quickly recover from unexpected challenges, is the assurance that the project can regain stability.

❖ **Continuous Integration:**

Connecting Git to a continuous integration tool like Jenkins makes the whole process of testing and deployment through automation very easy. When changes are moved to the repository, automated builds and tests are triggered, meaning that only correctly coded software is indeed the only one to go live.

Collaboration Tools Integration:

Git repositories are mainly integrated with - Slack, Microsoft Teams or similar tools. This includes getting the ability of sending timestamps, pull requests and builds to given users. The feature makes the communication of pull requests and statuses robust thereby allowing the team to work quickly and to keep a smooth communication among team members.

- **Documentation Management:**

Git repositories become one major source of information in documentation projects. Apart from the code used for the changes, the documentation can be versioned by Markdown files, READMEs, and a lot of other things, thus the documentation will be in line with the present state of the codebase.

- **Code Review:**

The maintenance of code quality, follow-up coding standards, and team members' knowledge sharing are the guarantees of this system so stand fast.

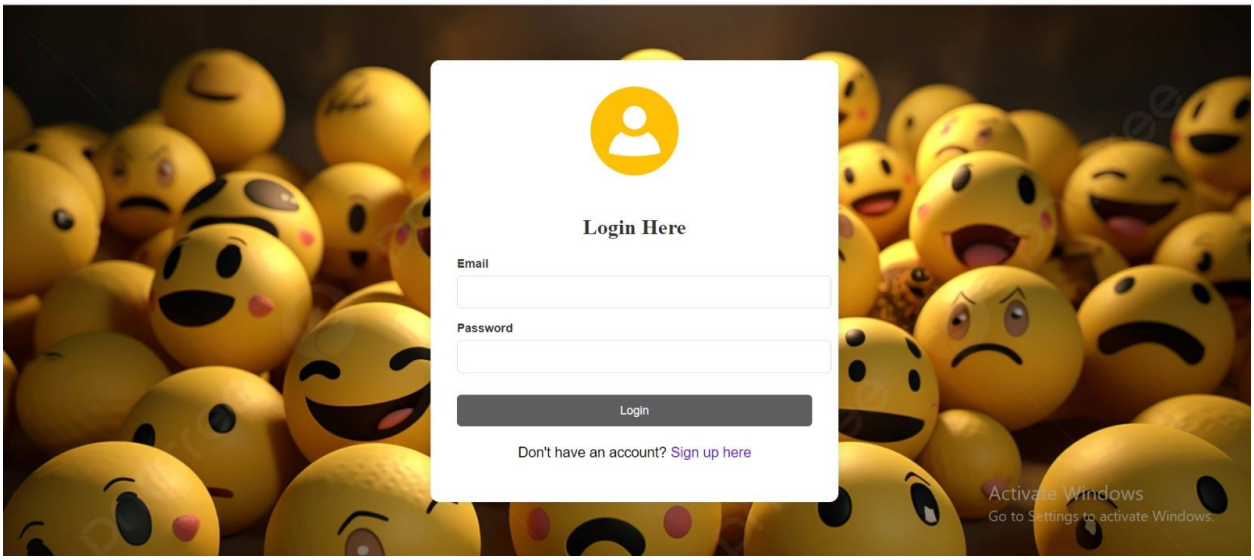


Figure 5.6.1 Login Page

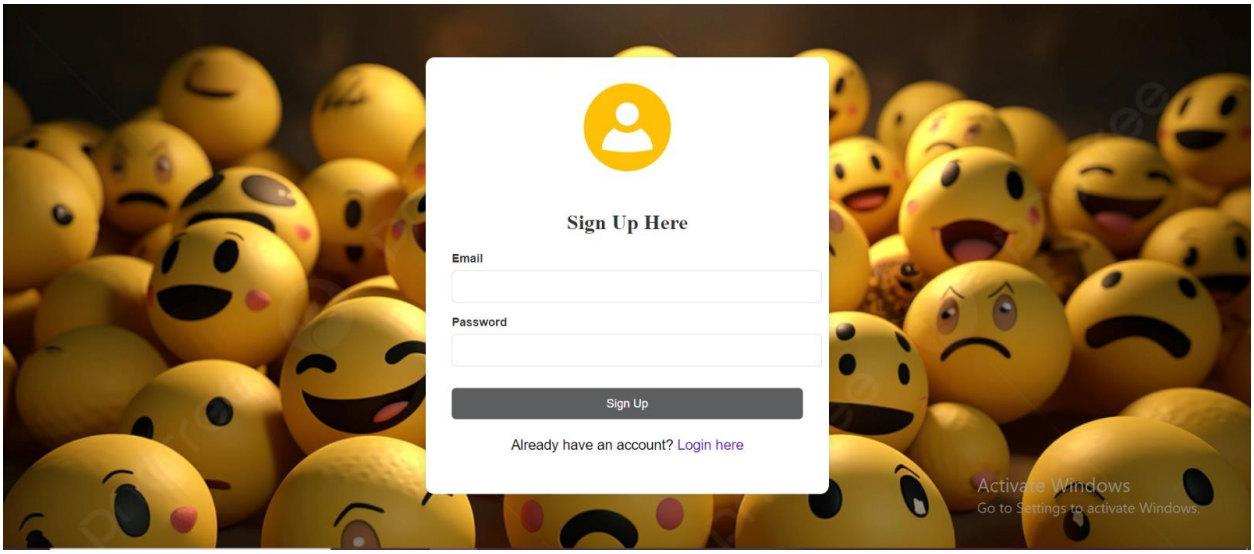


Figure 5.6.2 Sign up page

Figure 5.6.3 User interface

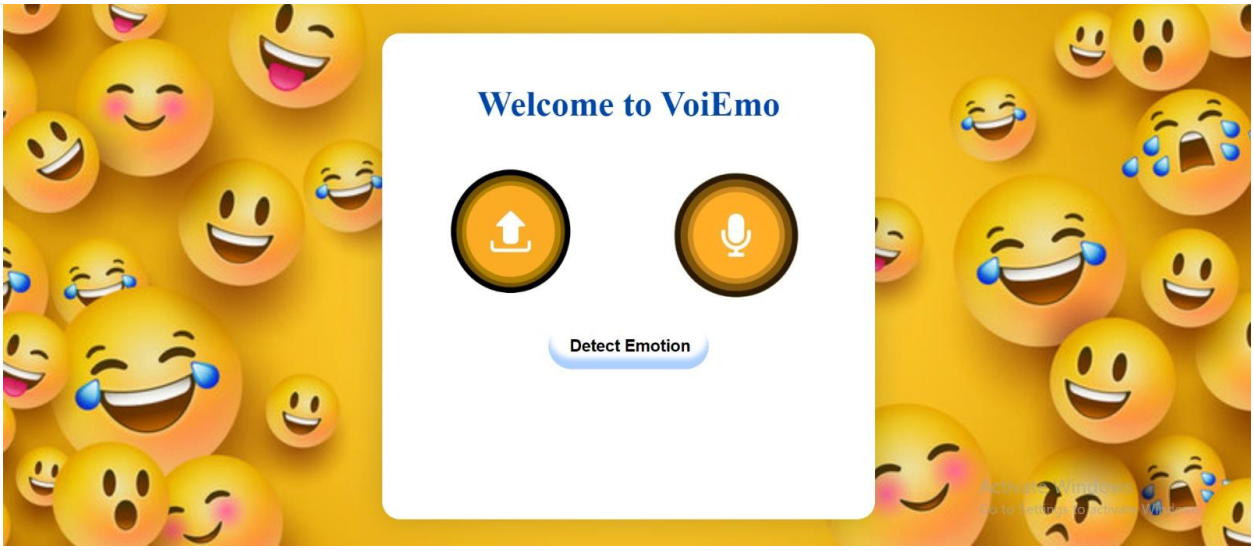


Figure 5.6.4 Interface

Chapter 6

Testing and Evaluation

Chapter 6: Testing and Evaluation

In this chapter we can verify the AI voice assistant system performance and functionality through extensive testing and evaluation. Also check problems and weaknesses and makes necessary improvements to improve the system's usability and efficacy.

6.1. Equivalence partitioning

The software testing, equivalency partitioning is a practice that categorizes the input space into the equivalence classes. It is expected that the system works the same, regardless of which class of inputs the system uses.

Voice Quality:

Crystal-clear: Voice modulation in inputs is distinct and transparent.

Muffled: Voice quality inputs that have a small amount of distortion.

Noise in the Background: Signals that contain interference or background noise.

Low volume: Low volume sources of input.

- **Language:**

  Standard English: Spoken inputs are in the English language standard.
- **Rate of Speech:**

  Normal Pace: The input is spoken at the typical tempo of a conversation.

Equivalence partitioning		
Invalid	**Valid**	**Invalid**
Not registered	Registered	**Unauthorized**
Result Status	**Pass**	**Fail**

Table 6. 1.1 Equivalence partitioning

Field Name	Invalid	Valid	Invalid
Email	Less than 11 characters	Characters 11-255 Email type	**More than 255** **Not email type**
Password	**Less than 6 characters**	**Characters 6-50**	**More than 50**

Table 6.1.2 Sign up Page

Field Name	Invalid	Valid	Invalid
Email	Less than 11 characters	Characters11- 255 Email type	**More than255** **Not email type**
Password	**Less than 6 characters**	**Characters 6-50**	**More than 50**

Table 6.1.3 Login Page

6.2. Boundary value analysis

Boundary Value Analysis, a worldwide known method of the AI Emotion Recognition Project, is the testing process in which one tests the behavior of a system at different input ranges. Besides fixing shortcomings the tests include speed, pitch, and volume, there is a framework by which this happens and the provider sets boundaries thus the responses have to come from these limits. The situations of error and the failures that lead to a wide band of user inputs are tested with some conditions that are unique.

Table 6.2 Boundary value analysis

	Password:	
INVALID **(min-1)**	**Valid** **(min, +min, -max, max)**	**INVALID** **(max+1)**
7	8 – 30	31

	Email:	
INVALID **(min-1)**	**Valid** **(min, +min,-max, max)**	**INVALID** **(max+1)**
15	16-30	**31**

6.3. Data flow testing

Here's an outline of potential data flow testing scenarios:

Input Data Flow Testing: In the input data flow testing, make sure which user voice entry patients were honest and correctly entered into the system. For it to be sustainable, verify it by testing several voice inputs like, for instance, the person feels happy, sad, or astonishing.

Table 6.3 Data flow testing

Data flow (testing)		
Data unit name:	Data flow method:	Test results:
Login	Click Login → Login Page open → Fill the form → And successfully go to home page	Successful
Register	Click on Register/ sign up page → Go to Sign up page → Fill the credentials → Successfully go to Login Page	Successful
Home	Now you are on your working page→ Buttons showed	Successful
Real time detection	Click on voice Button→ Get your voice be recorded→ And test your emotions on your real voice	Successful
Upload voices	Click on upload button→ Select the voice you want to test the emotion → Emotion detected on recorded data	Successful

Response Generation Data Flow Testing: Testing a conversational model response that is based on the user's voice input, which has just recognized them, is carried out by the response generation data flow testing. Making sure that the answers of your device are relevant to what the user has said and do not contradict the context.

Feedback Data Flow Testing: Check the feedback data flow system to capture the opinions of the users on the system's responses. Review how the system manages and uses the given information to improve its efficiency for the longer term within the process of responding to users according to their choices.

Integration Data Flow Testing: Find the correct arrangement of the voice input module, response generation module and the feedback processing module with respect to the data flow location and receive signals such as the controls and responses. In the very first passage a cross communication takes place among the components which enables it (communication) to actuate the communication, and operations are carried out smoothly.

Error Handling Data Flow Testing: Try out its capability to handle errors and exceptions specifically arising from data flow, i.e. unusual input formats, network drops, and errors in the processing stage.

Evaluate that the proper error message appears correct and the system enables to recover from fails without malfunctioning., and the system gracefully recovers from errors without compromising functionality.

6.4. Unit testing

We can also focus on unit testing :

- **Component Isolation**:

EmoBot Express is likely to interact with third-party services or APIs that are dedicated to the recognition of emotions. When we want to test the units apart, we create fake data which simulates the connections.

Unit tests concentrate on breaking the app into smaller parts. Every single part of it, such as login functionality, emotion recognition, or response generation, is tracked independently.

Let's talk about the login functionality that can be rather rigorously tested by entering of different email and password combinations and examining if the app reacts in the right way.

- **Response Validation**:

Unit tests verify that the responses align with expected emotional contexts.

- **Mocking External Services**:

There is a high probability that EmoBot Express sends requests to the outside sources for the analysis of the emotions of the persons' pictures. Developers write the unit tests using test doubles to pretend these requests are happening and study the Ap that processes these to get reliable results.

Table 6.4 Unit testing

Software Name: Unit	Method:	Test Results:
Login	Validation, Exception Handling, Tools & Events.	Successful.

Register	Validation, Exception Handling, Tools & Events.	Successful.
Home	Validation, Exception Handling, Tools & Events.	Successful.
Real time detection	Validation, Hardware Check, Exception Handling, Tools & Events.	Successful.
Upload voices	Validation, Exception Handling, Tools & Events.	Successful.

6.5. Integration testing

order to evaluate the voice assistant system, one has to take a closer look at the interrelation of the parts and their connection points to the whole system to know if the several components act in conjunction without any issues.

- Initiate the voice input processing module, and verify that it sends necessary information to the correct location. In this particular case, you must provide, via a set of scripts, many voice recordings of different sources and backgrounds. Then the algorithm will have to detect and categorize the speech parts that are differentiable for each of them so that it can bring about a profitable change in some way through the discrimination that has been implemented.

- · Keep a check on the voice detection module's integration with the system response generation system. Thus, we will be able to verify that the adequate system's responses are appropriately situated and that they are correct in the user's input case as directed by the algorithm.

- you can also inspect the mobile app side of the question in terms of the React Native framework. This means user testing the app, for instance, to speak commands and receive responses, thus, to maintain the easy interrelation of the front end with the AI back end.
- At the same time, we would be considering edge cases (situation where single program ends or more input) and how the system would respond. During integration testing, you can also simulate a variety of scenarios and edge cases to verify that-zap the whole system functions correctly under different conditions

TABLE 6.5 Integration testing

Software unit Name:	Test unit with:	Testing Method:	Test Results:
Login	Login page	Validation, Exception Handling, Tools & Events.	Successful.
Register	Register page	Validation, Exception Handling, Tools & Events.	Successful.
Home	Show the home page	Validation, Exception Handling, Tools & Events.	Successful.
Real time detection	Record your real time voice	Validation, Hardware Check, Exception Handling, Tools & Events.	Successful.
Upload voices	Upload the recorded voice you want to check the emotion of.	Validation, Exception Handling, Tools & Events.	Successful.

6.6. Performance testing

The test should provide proof that the AI voice assistant produces a high-end guarantee to the final user with a quicker response, even the highest load levels.

The main idea of AI Voice Assistant is to ensure its agility in multiple types of contexts through performance testing. The examination is oriented to a question of how the system may act in real conditions-in brief, this is the question of robustness and the efficacy of such a system that can cooperate with users without losing their time and energy.
The components tested include different metrics. One of the criteria that the system must meet is the response time, which is the time it takes to process and prompt a user's query.
This AI system has an amazing property, among them there are also metrics for resource utilization and performance (e.g. memory and CPU usage) to measure how much work is done by a single program and how the system resources are affected.
Running through every performance indicator is necessary for the AI Voice Assistant to be able to immediately respond to the user thereby establishing a high-quality user experience even when the system is at its maximum.

Table 6.6 performance testing

Sr no.#	Function Name	Performance
1	UI loading	High
2	Connection of modules	High
3	Storing data	High
4	Retrieving data	High

6.7. Stress Testing

We evaluate the AI Voice Assistant system's scalability by placing it under extreme stress and monitoring its response to see how it handles the worst-case scenarios. To test how many and how quickly a device can handle voice inputs, we first prod it with a stick by delivering a large number of simultaneous speech inputs via a selection of devices, roughly representing worst-case usage patterns.

We introduce errors into the system, mostly in the form of corrupt speech data or network outages, and watch as it functions under the failing circumstances before recovering without compromising its functionality. Tolerance for Faults

Chapter 7

Summary, Conclusion and Future Enhancements

Chapter 7: Summary, Conclusion & Future Enhancements

7.1. Project Summary

About the Project this project is a Real-time emotion Detector and Voice based AI Voice Assistant system which is developed using React Native Mobile application. The system can detect diverse categories of emotions, through state-of-the-art emotion detection algorithms that detect emotions such as happiness, sadness, anger, among others, each speaker can speak into the system in a natural way and receive target responses based on their emotion [sic each speaker and the statement.This could help in increasing the user engagement and help the project in enhancing user satisfaction by offering conversational experiences powered by personal and empathetic conversations.

7.2. Achievements and Improvements

In section of 'Features & Achievements', there is a record of the major tech advances and outcomes discovered in the imaging and machine intelligence systems project. This was a tricky thing to do since it needs identifying multiple emotions in voice signals for the system detecting emotions in real time to react but also a lot of good user feedbacks and be happy with the system.

7.3. Critical Review

An in-depth review of the AI voice assistant project is comprised in the Critical Review section. This paper evaluates its strengths and weaknesses by examining its technology implementation and expression detection in images and videos against its performance in real-time interactions. Additionally, it explores the project within the existing backbone of AI-driven systems and technological progress. While these findings may suggest some of the opportunities that could be leveraged for improvement (e.g., to improve accuracy in emotion detection, to increase persuasiveness or conversational principles capability to meet user requests in a better way...) this critical assessment remains high level. It shows progress in AI-driven voice assistants while leaving room for further evolution.

7.4. Lessons Learnt

Working on the development of an artificial intelligence voice assistant system for my final year project gave me insights into a number of significant topics. The difficulty of putting machine learning algorithms for voice recognition of emotions into practice was one important lesson discovered. In addition, I developed expertise incorporating intelligence technology into practical applications, especially in the area of developing React Native mobile apps. The project emphasized how important user-centered design is. emphasized how crucial it is to recognize and respond to consumers' emotional states in order to improve their experience. Overall, this project helped me to better grasp AI-driven system development and how it affects technology.

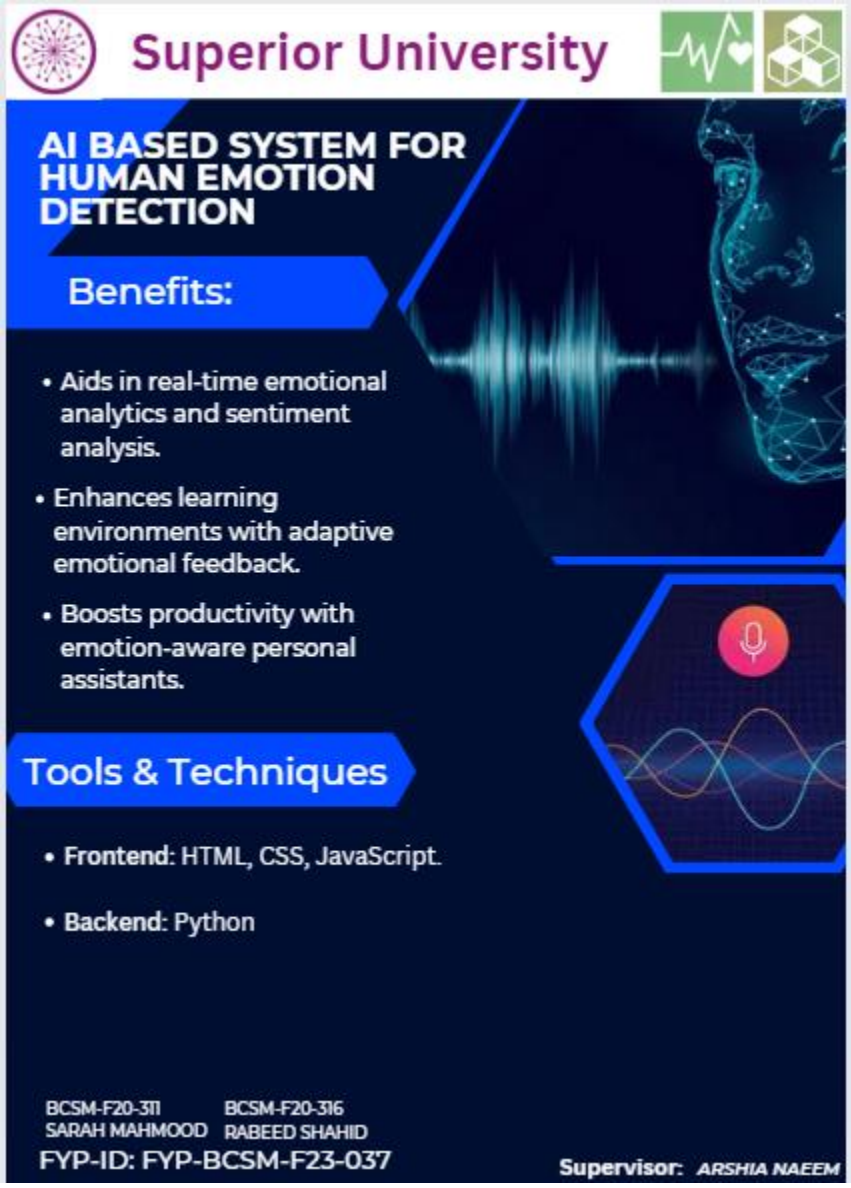
7.5. Future Enhancements/Recommendations

Customized user profiles that adjust replies based on a person's preferences, mood, and emotions may be included in these next updates. The technology could possibly handle other languages by incorporating machine learning algorithms to improve emotion-detection methods. Adding vocal synthesis functionality to deliver more natural and expressive responses different integrations, such as with social media or smart homes to expand its reach even more; further escalation a need of the hour AI assistant to users in all sorts of daily settings.

Appendices

Appendix C: Information / Promotional Material

C.1. Standee



Reference and Bibliography

Reference and Bibliography

1. Koolagudi, S.G. and Rao, K.S. (2012). Emotion recognition from speech: a review. International Journal of Speech Technology, 15(2), pp.99-117. https://link.springer.com/article/10.1007/s10772-011-9125-1

2. Lugger, M. and Yang, B. (2008). The relevance of voice quality features in speaker independent emotion recognition. 2008 IEEE International Conference on Acoustics, Speech and Signal Processing, pp.5101-5104. https://researchr.org/publication/icassp-2008.

3. Mehmood, R., Ahmad, J., & Rehman, S. U. (2020). Deep learning-based speech emotion recognition: State-of-the-art review. IEEE Access, 8, 99545-99563. https://scholar.archive.org/work/43npoaavzbhtfnmy7mhizc3k3m/access/wayback/https://ieeexplore.ieee.org/ielx7/6287639/6514899/09226478.pdf

4. Gideon, J., & Liu, H. (2020). Emotion recognition from speech using deep learning techniques: A comprehensive review. Journal of Ambient Intelligence and Humanized Computing, 11, 135-149.

5. https://www.researchgate.net/publication/335360469_Speech_Emotion_Recognition_Using_Deep_Learning_Techniques_A_Review.

6. Alam, M. R., Murad, M. A. R., Khan, S., & Rashid, M. M. (2021). Speech emotion recognition using machine learning and deep learning techniques: A systematic review. Neural Computing and Applications, 33, 11083–11104. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10049636/